



XAT 2013 Question Paper

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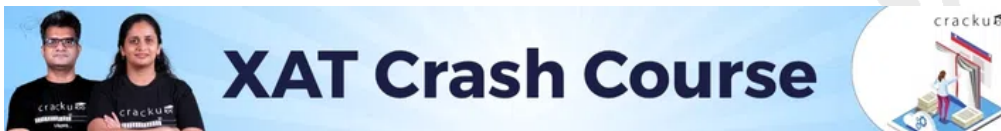
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Verbal and Logical ability

1. Fill in the blanks with the most appropriate option that follows:

..... wolf, meeting withlamb astray fromfold, resolved not to lay violent hands on him, but to find some plea to justify to lamb wolf's right to eat him.

- A a, a, the, the, the
- B the, a, the, a, the
- C a, a, a, the, the
- D the, the, the, the, the
- E the, a, the, a, a



2. bat who fell upon..... ground and was caught by..... weasel pleaded to be spared his life. weasel refused, saying that he was by nature enemy of all birds. bat assured him that he was not bird, but mouse, and thus was set free.

- A a, the, a, the, the, the, the, a
- B a, the, a, the, a, the, a, a
- C the, a, a, the, the, a, the, the
- D a, the, a, the, the, the, a, a
- E the, a, a, a, the, the, a, a

3. He got next morning, to be sure, and had his meals usual, though he ate and had more, I am afraid, than his usual supply of rum, for he helped himself..... the bar, scowling and blowing..... his nose, and no one dared..... cross him.

- A Down, like, a little, out of, out, to
- B Down, as, little, of, out, to
- C Downstairs, as, little, out of, out of, through
- D Downstairs, like, a little, out, of, to
- E Down, like, a little, of, of through

4. Choose the most appropriate option after reading the following statements.

1. Whether due to haste or design, the new laws are marked by vagueness, leaving officials all down the organization's bureaucratic chain great latitude in enforcing them.
2. The opacity of the language leaves the law open to manipulation on political grounds.

- A Statement 2 can be induced from statement 1.
- B Statement 1 can be induced from statement 2.

- C Statement 2 can be deduced from statement 1.
- D Statement 1 can be deduced from statement 2.
- E Statements 1 and 2 are independent.



5. Choose the most appropriate option after reading the following statements.
1. If there is any endeavour whose fruits should be freely available, that endeavour is surely publicly financed science.
 2. There is a widespread feeling that the journal publishers who have mediated the scientific exchange for the past century or more are becoming an impediment to free distribution of knowledge.
 3. Internet revolution is happening, making knowledge transfer cheaper. Technology permits it; researchers and politicians want it, more public money can be spent on it.
- A Statement 2 definitely illustrates statement 1.
 - B Statement 3 is a facilitating condition for statement 1
 - C Statement 3 states a condition under which statement 1 would be invalid
 - D Statement 2 can be deduced from statement 3 but independent of Statement 1.
 - E Statement 1, 2 and 3 are necessarily independent.
6. Choose the most appropriate option after reading the following statements.
1. Business schools are ideally positioned to point out when an action that provides a benefit for an individual comes at a cost to society, but in reality they rarely bother.
 2. It is part of the malaise that has befallen the political debate on capitalism, which has been taken over by special interests and people who have no faith in a real market-based system.
 3. When governments favours the private sector it is all too often by being "pro-business" rather than "pro-market", meaning that favourable conditions are provided to particular institutions rather than to institutions broadly.
- A Statements 1 and 2 are necessarily dependent.
 - B Statements 2 and 3 are necessarily dependent.
 - C Statements 2 and 3 may be dependent.
 - D Statements 1, 2 and 3 cannot be independent.
 - E All the three statements are necessarily independent.
7. Read the following sentence and choose the best alternative which should replace the italicized part of the sentence.
- To be a great manager *requires, strong inter-personal skills, the ability to think fast, and* demands a can-do attitude.
- A requires strong inter-personal skills, the ability to think fast, and demands
 - B requires strong inter-personal skills, the ability to think fast, and
 - C requires strong inter-personal skills, demands the ability to think fast, and

- D requiring strong inter-personal skill, an ability to think fast, and demands
- E demands strong inter-personal skills, an ability to think fast, but with



8. Read the following sentence and choose the best alternative which should replace the italicized part of the sentence.

The tremendous insight of Einstein was that the passage of time does not appear to be the same *while standing still as it does to a person traveling at a speed* which is a significant fraction of the speed of light.

- A** while standing still as it does to a person traveling
- B** to a person standing still as to a person traveling
- C** to a person who is standing still as a person who is traveling
- D** while standing still as to traveling
- E** to a person standing still as to a person who travels

9. Read the following sentence and choose the best alternative which should replace the italicized part of the sentence.

Economic theory fails to explain the extent *to which savings from personal income has Shifted* to short-term bonds, money-market funds, and other near-term investments by the instability in the futures market.

- A** to which savings from personal income has shifted
- B** of savings from personal income that has been shifted
- C** of savings from personal income shifting
- D** to which savings from personal income have shifted
- E** to which savings from personal income have been shifted

- 10.** Read the following sentences and choose the option that best arranges them in a logical order.

1. I was scarcely in position ere my enemies began to arrive, seven or eight of them, running hard, their feet beating out of time along the road and the man with the lantern some paces in front.
2. My curiosity, in a sense, was stronger than my fear, for I could not remain where I was, but crept back to the bank again, whence, sheltering my head behind a bush of broom, I might command the road before our door.
3. Three men ran together, hand in hand; and I made out, even through the mist, that the middle man of this trio was the blind beggar.
4. The next moment his voice showed me that I was right.

- A** 1,2,3, 4
B 2, 1, 3, 4
C 1,2,4,3
D 1,3,4,2
E 1,4,2,3

11. Read the following sentences and choose the option that best arranges them in a logical order.
1. Finally he took a wrong turn and ran a few, steps past me, towards the hamlet, crying, "Johnny, Black Dog, Dirk," and other names, "you won't leave old Pew, mates—not old Pew!"
 2. This quarrel was the saving of us, for while it was still raging, another sound came from the top of the hill on the side of the hamlet—the tramp of horses galloping.
 3. And that was plainly the last signal of danger, for the buccaneers turned at once and ran, separating in every direction, one seaward along the cove, one slant across the hill, and so on, so that in half a minute not a sign of them remained but Pew.
 4. Him they had deserted, whether in sheer panic or out of revenge for his ill words and blows I know not; but there he remained behind, tapping up and down the road in a frenzy, and groping and calling for his comrades.
 5. Almost at the same time a pistol-shot, flash and report, came from the hedge side.
- A 5, 1, 3, 4, 2
B 1, 2, 3, 5, 4
C 2, 5, 3, 4, 1
D 4, 3, 2, 5, 1
E 2, 5, 4, 3, 1
12. Read the following sentences and choose the option that best arranges them in a logical order.
1. As chroniclers of an incremental process, they discover that additional research makes it harder, not easier, to answer questions like: When was oxygen discovered? Who first conceived of energy conservation?
 2. Simultaneously, these same historians confront growing difficulties in distinguishing the "scientific" component of past observation and belief from what their predecessors had readily labeled "error" and "superstition."
 3. Increasingly, a few of them suspect that these are simply the wrong sorts of questions to ask. Perhaps science does not develop by the accumulation of individual discoveries and inventions.
 4. In recent years, however, a few historians of science have been finding it more and more difficult to fulfill the functions that the concept of development-by-accumulation assigns to them.
- A 2,1,3,4
B 4,3,1,2
C 4,2,3,1
D 4,3,2,1
E 4,1,3,2

Instructions [13 - 16]

Analyse the following passage and provide appropriate answers

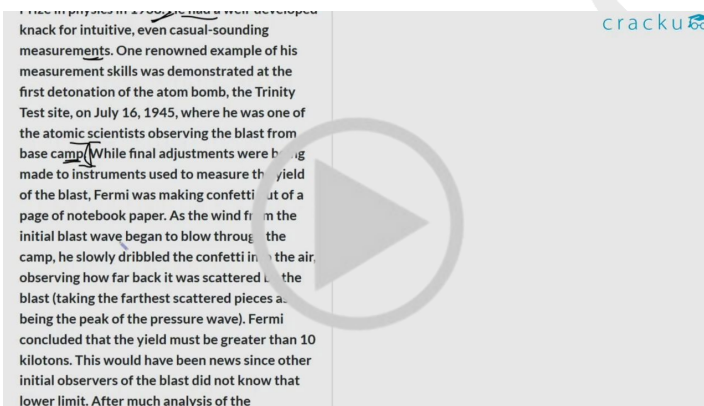
An example of a scientist who could measure without instruments is Enrico Fermi (1901-1954), a physicist who won the Nobel Prize in physics in 1938. He had a well-developed knack for intuitive, even casual-sounding measurements. One renowned example of his measurement skills was demonstrated at the first detonation of the atom bomb, the Trinity Test site, on July 16, 1945, where he was one of the atomic scientists observing the blast from base camp. While final adjustments were being made to instruments used to measure the yield of

the blast, Fermi was making confetti out of a page of notebook paper. As the wind from the initial blast wave began to blow through the camp, he slowly dribbled the confetti into the air, observing how far back it was scattered by the blast (taking the farthest scattered pieces as being the peak of the pressure wave). Fermi concluded that the yield must be greater than 10 kilotons. This would have been news since other initial observers of the blast did not know that lower limit. After much analysis of the instrument readings, the final yield estimate was determined to be 18.6 kilotons. Like Eratosthenes, Fermi was aware of a rule relating one simple observation—the scattering of confetti in the wind—to a quantity he wanted to measure.

The value of quick estimates was something Fermi was familiar with throughout his career. He was famous for teaching his students skills at approximation of fanciful-sounding quantities that, at first glance, they might presume they knew nothing about. The best-known example of such a "Fermi question" was Fermi asking his students to estimate the number of piano tuners in Chicago, when no one knows the answer. His students—science and engineering majors—would begin by saying that they could not possibly know anything about such a quantity. Of course, some solutions would be to simply do a count of every piano tuner perhaps by looking up advertisements, checking with a licensing agency of some sort, and so on. But Fermi was trying to teach his students how to solve problems where the ability to confirm the results would not be so easy. He wanted them to figure out that they knew something about the quantity in question.

13. Suppose you apply the same logic as Fermi applied to confetti, which of the following statements would be the most appropriate?

- A** You can calculate the minimum pressure inside the cooker by calculating the maximum distance travelled by any of its parts after it explodes.
- B** You can calculate the average potency of a firecracker by calculating the distance covered by one of its bigger fragments.
- C** You can easily find out the average potency of an earthquake by measuring the length of a crack it makes on the surface of the earth.
- D** You can calculate the exact volume of water stored in a tank by measuring the distance covered by the stream of water coming out of the tap fixed on the lower corner of the tank.
- E** All of the above conclusions can be drawn.

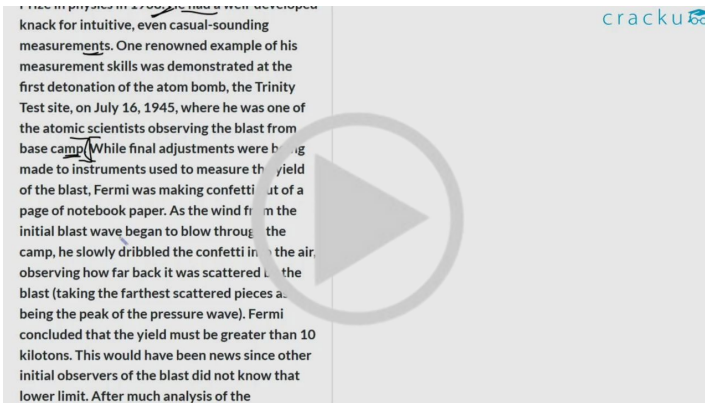


 VIDEO SOLUTION



14. Quick estimate, as per Fermi, is most useful in:

- A In finding an approximate that is more useful than existing values.
- B In finding out the exact minimum value of an estimate
- C In finding out the exact maximum value of an estimate
- D In finding out the range of values of an estimate
- E In finding out the average value of an estimate



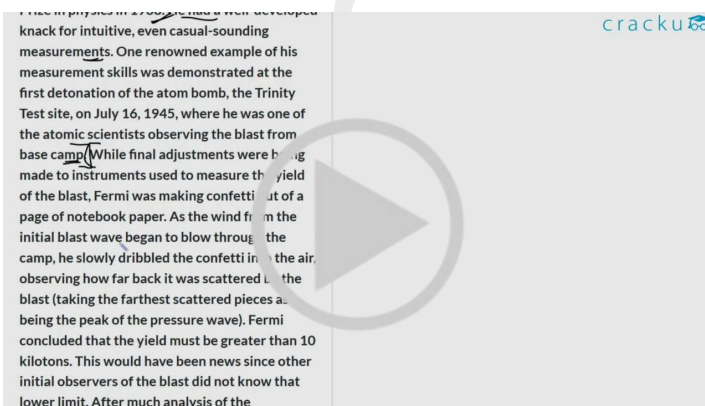
VIDEO SOLUTION

15. Given below are some statements that attempt to capture the central idea of the passage:

1. It is useful to estimate; even when the exact answer is known.
2. It is possible to estimate any physical quantity.
3. It is possible to estimate the number of units of a newly launched car that can be sold in a city.
4. Fermi was a genius.

Which of the following statements best captures the central idea?

- A 1, 2 and 4
- B 2, 3 and 4
- C 2 and 3
- D 2 only
- E 1, 2 and 3

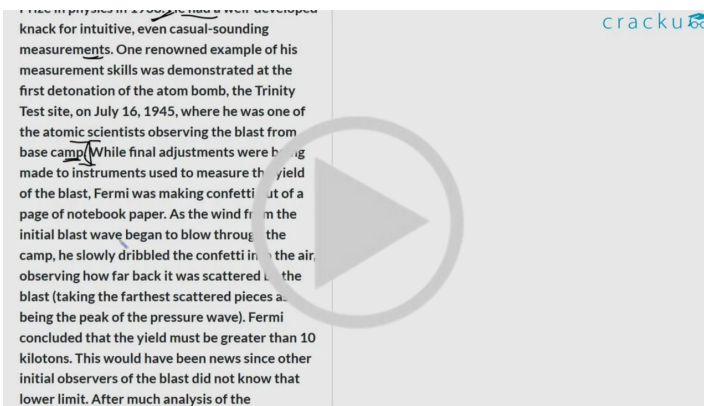


16. Read the statements given below:

1. Atomic bomb detonation was a result of Fermi's Nobel Prize contribution
2. Fermi's students respected him as a scientist
3. Yield of atomic bomb can only be measured in Kilotons

Which of the following statement(s) can be inferred from the passage?

- A 1, 2
- B 2, 3
- C 1, 3
- D 2 only
- E None of the three statements is correct, hence cannot be inferred from the passage.



XAT Syllabus PDF



Instructions [17 - 20]

Analyse the following passage and provide appropriate answers for the follow.

Popper claimed, scientific beliefs are universal in character, and have to be so if they are to serve us in explanation and prediction. For the universality of a scientific belief implies that, no matter how many instances we have found positive, there will always be an indefinite number of unexamined instances which may or may not also be positive. We have no good reason for supposing that any of these unexamined instances will be positive, or will be negative, so we must refrain from drawing any conclusions. On the other hand, a single negative instance is sufficient to prove that the belief is false, for such an instance is logically incompatible with the universal truth of the belief. Provided, therefore, that the instance is accepted as negative we must conclude that the scientific belief is false. In short, we can sometimes deduce that a universal scientific belief is false but we can never induce that a universal scientific belief is true.

It is sometimes argued that this 'asymmetry' between verification and falsification is not nearly as pronounced as Popper declared it to be. Thus, there is no inconsistency in holding that a universal scientific belief is false despite any number of positive instances; and there is no inconsistency either in holding that a universal scientific belief is true despite the evidence of a negative instance. For the belief that an instance is negative is

itself a scientific belief and may be falsified by experimental evidence which we accept and which is inconsistent with it. When, for example, we draw a right-angled triangle on the surface of a sphere using parts of three great circles for its sides, and discover that for this triangle Pythagoras' Theorem does not hold, we may decide that this apparently negative instance is not really negative because it is not a genuine instance at all. Triangles drawn on the surfaces of spheres are not the sort of triangles which fall within the scope of Pythagoras' Theorem. Falsification, that is to say, is no more capable of yielding conclusive rejections of scientific belief than verification is of yielding conclusive acceptances of scientific beliefs. The asymmetry between falsification and verification, therefore, has less logical significance than Popper supposed.

We should, though, resist this reasoning. Falsifications may not be conclusive, for the acceptances on which rejections are based are always provisional acceptances. But, nevertheless, it remains the case that, in falsification, if we accept falsifying claims then, to remain consistent, we must reject falsified claims. On the other hand, although verifications are also not conclusive, our acceptance or rejection of verifying instances has no implications concerning the acceptance or rejection of verified claims. Falsifying claims sometimes give us a good reason for rejecting a scientific belief, namely when the claims are accepted. But verifying claims, even when accepted, give us no good and appropriate reason for accepting any scientific belief, because any such reason would have to be inductive to be appropriate and there are no good inductive reasons.

17. According to Popper, the statement "Scientific beliefs are universal in character" implies that

- A** Positive instances of scientific belief imply that it is universal in character.
- B** There are equal numbers of negative and positive instances of a universal scientific belief.
- C** If there are negative and positive instances of a scientific belief then it cannot be universal.
- D** We can only deduce that a scientific belief is false but cannot induce that it is true.
- E** We can only induce that a scientific belief is false but cannot induce that it is true.

18. The statement, "this 'asymmetry' between verification and falsification is not nearly as pronounced as Popper declared it to be", implies that

- A** Falsification is better than verification in universal acceptance of scientific beliefs.
- B** Verification is better than falsification in universal acceptance of scientific beliefs.
- C** Both falsification and verification together can result in universal acceptance of scientific beliefs.
- D** Capability of falsification in accepting of scientific beliefs is not better than that of verification in rejection of scientific beliefs.
- E** Capability of falsification in rejection of scientific beliefs is not always better than that of verification in acceptance of scientific beliefs.

19. With which of the following statements, would the author agree most?

- A** Verification is better than falsification in establishing the claims.
- B** Falsification and verification are equally good in establishing the claims.
- C** Verification and falsification are equally bad in establishing the claims.
- D** Falsification is better than verification in disproving the claims.
- E** Verification is better than falsification in disproving the claims.



20. Which of the following would be the most appropriate conclusion?

- A Falsification gives us an appropriate reason for rejecting a scientific belief.
- B Falsification gives us all the reasons for accepting a claim.
- C Verification gives us a reason for rejecting a claim.
- D Verification gives us an appropriate reason for accepting a scientific belief.
- E Verification gives us an appropriate reason for rejecting a scientific belief.

Instructions [21 - 25]

Analyse the following passage and provide appropriate answers for the through that follow.

Soros, we must note, has never been a champion of free market capitalism. He has followed for nearly all his public life the political ideas of the late Sir Karl Popper who laid out a rather jumbled case for what he dubbed "the open society" in his *The Open Society and Its Enemies* (1953). Such a society is what we ordinarily call the pragmatic system in which politicians get involved in people's lives but without any heavy theoretical machinery to guide them, simply as the ad hoc parental authorities who are believed to be needed to keep us all on the straight and narrow. Popper was at one time a Marxist socialist but became disillusioned with that idea because he came to believe that systematic ideas do not work in any area of human concern.

The Popperian open society Soros promotes is characterized by a very general policy of having no firm principles, not even those needed for it to have some constancy and integrity. This makes the open society a rather wobbly idea, since even what Popper himself regarded as central to all human thinking, critical rationalism, may be undermined by the openness of the open society since its main target is negative avoid dogmatic thinking, and avoid anything that even comes close to a set of unbreachable principles. No, the open society is open to anything at all, at least for experimental purposes. No holds are barred, which, if you think about it, undermines even that very idea and becomes unworkable.

Accordingly, in a society Soros regards suited to human community living, the state can manipulate many aspects of human life, including, of course; the economic behavior of individuals and firms. It can control the money supply, impose wage and price controls, dabble in demand or supply-side economics, and do nearly everything a central planning board might —provided it does not settle into any one policy firmly, unbendingly. That is the gist of Soros's Popperian politics.

Soros' distrusts capitalism in particular, because of the alleged inadequacy of neoclassical economics, the technical economic underpinnings of capitalist thinking offered up in many university economics departments. He, like many others outside and even inside the economics discipline, finds the arid reductionism of this social science false to the facts, and rightly so. But the defense of capitalist free markets does not rest on this position.

Neo-classical thinking depends in large part on the 18th- and 19th-century belief that human society operates according to laws, not unlike those that govern the physical universe. Most of social science embraced that faith, so economics isn't unusual in its loyalty to classical mechanics. Nor do all economists take the deterministic lawfulness of economic science literally — some understand that the laws begin to operate only once people embark upon economic pursuits. Outside their commercial ventures, people can follow different principles and priorities, even if it is undeniable that most of their endeavors have economic features. Yet, it would be foolish to construe religion or romance or even scientific inquiry as solely explicable by reference to the laws of economics.

In his criticism of neo-classical economic science, then, George Soros has a point: the discipline is too dependent on Newtonian physics as the model of science. As a result, the predictions of economists who look at markets as if they were machines need to be taken with a grain of salt. Some — for example the school of Austrian economists — have made exactly that point against the neo-classical.

Soros draws a mistaken inference: if one defense of the market is flawed, the market lacks defense. This is wrong. If it is true that from A we can infer B, it does not prove that B can only be inferred from A; C or Z, too, might be a reason for B.

21. As per the paragraph, author believes that

- A Free market capitalism can be explained using neo-classical economics.
- B Neo-classical economics does not address the idea of free-market system.
- C Free market capitalism and open society are not different from each other.
- D Free market capitalism and laissez-faire are not different from each other.
- E Technical underpinning of neo-classical economics can address the idea of laissez-faire.

22. As per the paragraph, which of the following statements is true?

- A Economic benefits of open society and laissez-faire are same.
- B Soros' open society means no interference from the government.
- C Free market capitalism means no interference from the government.
- D Laws of economics are not capable of explaining the human nature completely.
- E Laws of economics capture the human nature completely as most of the human endeavors are economic in nature.



23. According to the author,

- A George Soros believes in regulated economies.
- B George Soros does not believe in government intervention in state policies.
- C George Soros believes in state intervention provided it does not remain static.
- D George Soros believes that laissez-faire economies perform better than free-market economies.
- E George Soros believes that free-market economies perform better than controlled economies.

24. According to the author which of the following statement could be true about critical rationalism.

- A Ideas of critical rationalism underpin the foundation of neo-classical economics.
- B Ideas of critical rationalism underpin the foundation of laissez-faire.
- C Ideas of critical rationalism underpin the foundation of open society.

- D Ideas of critical rationalism underpin the foundation of Newtonian physics.
- E None of the above.

25. The word deterministic (used in fourth line of fifth paragraph), in the above passage refers to:

- A An effect can only be caused by a single event.
- B An effect may be produced by many causes.
- C An effect cannot be produced by a cause.
- D Cause(s) of an effect can always be known.
- E Economics does not follow cause and effect relationship.



Decision Making

Instructions [26 - 28]

Marathe is a Vice President in a construction equipment company in the city of Mumbai. One day, his subordinate Bhonsle requested that Kale, a project manager, be transferred to the Chennai office from the Mumbai office. In Chennai, Kale would work alone as a researcher. Bhonsle gave the following reasons for his request: "Kale is known to frequently fight with his colleagues. Kale is conscientious and dedicated only when working alone. He is friendly with seniors, but refuses to work with colleagues, in a team. He cannot accept criticism and feels hostile and rejected. He is over-bearing and is generally a bad influence on the team."

Marathe called upon Gore, another project manager, and sought further information on Kale. Gore recalled that a former colleague, Lakhote (who was also Kale's former boss) had made a few remarks on his appraisal report about Kale. In his opinion, Kale was not fit for further promotion as he was emotionally unstable to work in groups though he had seven years of work experience. Lakhote had described Kale as too authoritative to work under anyone. Lakhote had further told Gore that Kale had an ailing wife, and an old mother, who does not want to stay with his wife.

26. Consider the following solutions to the problem mentioned above:

1. Marathe should transfer Kale to Chennai office
2. Marathe should try and verify the facts from other sources as well
3. Kale should be sacked
4. Kale should be demoted
5. Marathe should suggest Kale to visit a family counselor

Which of the following would be the most appropriate sequence of decisions in terms of immediacy: starting from immediate to a longer term solution?

- A 2, 1, 5
- B 1, 4, 2
- C 2, 3, 4
- D 2, 5, 1
- E 2, 5, 4

cracku  his opinion, Kale was not fit for further promotion as he was emotionally unstable to work in groups though he had seven years of work experience. Lakhote had described Kale as too authoritative to work under anyone. Lakhote had further told Gore that Kale had an ailing wife, and an old mother, who does not want to stay with his wife.

10 years of experience.

C) In the latest issue of a reputed journal, Xaviers Business Review, it was published that most top managers find it difficult to work in a group.

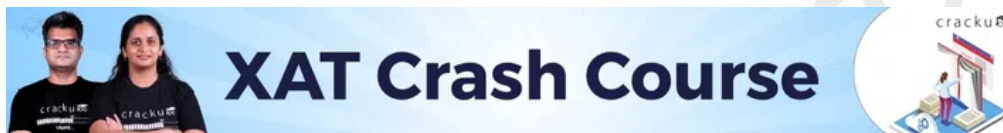
D) It was published in Xaviers Management Review (another reputed journal) that individuals who cannot work in teams find it difficult to adjust to a new location.

E) Bhonsle was of the opinion that emotionally unstable persons should not be promoted back to their original work life.

Kale
In personal
→ Chennai



 VIDEO SOLUTION



27. Consider the following solutions to the problem mentioned above:

1. Marathe should transfer Kale to Chennai office
2. Marathe should try and verify the facts from other sources as well
3. Kale should be sacked
4. Kale should be demoted
5. Marathe should suggest Kale to visit a family counselor

Marathe sought an appointment with Lakhote to find out ways to help Kale. Lakhote is of the opinion that the company's responsibility is restricted to the workplace and it should not try to address the personal problems of employees. If Marathe has to agree to Lakhote's opinion, which of the solutions presented in the previous question would be weakened:

- A 1
- B 2
- C 3
- D 4
- E 5

cracku  his opinion, Kale was not fit for further promotion as he was emotionally unstable to work in groups though he had seven years of work experience. Lakhote had described Kale as too authoritative to work under anyone. Lakhote had further told Gore that Kale had an ailing wife, and an old mother, who does not want to stay with his wife.

10 years of experience.

C) In the latest issue of a reputed journal, Xaviers Business Review, it was published that most top managers find it difficult to work in a group.

D) It was published in Xaviers Management Review (another reputed journal) that individuals who cannot work in teams find it difficult to adjust to a new location.

E) Bhonsle was of the opinion that emotionally unstable persons should not be promoted back to their original work life.

Kale
In personal
→ Chennai



 VIDEO SOLUTION

28. Consider the following solutions to the problem mentioned above:

1. Marathe should transfer Kale to Chennai office
2. Marathe should try and verify the facts from other sources as well
3. Kale should be sacked
4. Kale should be demoted
5. Marathe should suggest Kale to visit a family counselor

Which of the following statements, if true, would weaken the decision to sack Kale the most?

- A A Government of India study established that employees with 5-10 years of work experience tend to have conflicting responsibilities at home and office. However, these conflicts wither away after 10 years of experience.
- B Another article published in the magazine, Xaviers Quarterly, highlighted that employees' problems at home affect their performance at work
- C In the latest issue of a reputed journal, Xaviers Business Review, it was published that most top managers find it difficult to work in a group.
- D It was published in Xaviers Management Review (another reputed journal) that individuals who cannot work in teams find it difficult to adjust to a new location.
- E Bhonsle was of the opinion that emotionally unstable persons find it difficult to get back to normal working life.

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E) Bhonsle was of the opinion that emotionally unstable persons find it difficult to get back to normal working life.

Kale

In responsible

→ Confrontation

VIDEO SOLUTION

Instructions [29 - 31]

Read the following caselet and choose the best alternative:

It was the end of performance review cycle for the year 2012 when you asked your subordinates about any problems they were facing. Natrajan told you that an important member of his team, Vardarajan, who had also won the best performance award for the year 2011, was not taking interest in work. Despite Natrajan's counseling, no change was noticed in Vardarajan, rather his attitude deteriorated. You had also received such information from other employees. You had not interfered hoping that Natrajan, an experienced hand, would be able to solve the problem. But now that Natrajan himself brought this to your notice, you decided to call Vardarajan and counsel him.

29. Which of the following could be the most unlikely reason for Vardarajan's declining involvement in the workplace?

- A Vardarajan does not find the work challenging enough as he has already achieved the best performance award.
- B Others in the organization have been trying to pull him down, since he was declared best performer.
- C Vardarajan was not promoted after his superlative performance, while another colleague, Sundararajan, was promoted although he was not as good a performer as Vardarajan.
- D After putting in lots of effort for the superlative performance, Vardarajan felt burnt out.
- E Vardarajan was appreciated by his bossès for his achievement last year.



30. Vardarajan did not find his work challenging enough, given below are some steps that could be taken to motivate him:

- 1) Give Vardarajan a more challenging assignment
- 2) Transfer Vardarajan from Projects department to Training department
- 3) Ask him to take a vacation for two months
- 4) Send him for further training on decision making under stress

Which of the following conditions can be appropriate?

- A 1,2,3
- B 1,3,4
- C 1,4
- D 2,3,4

31. You overheard a conversation between Vardarajan and his colleague over an official dinner. He expressed his unhappiness about the fact that good performers were not given their due credit while poor performers were promoted faster. If Vardarajan is right, which of the following steps would help in creating a better organization?

- A Promote Vardarajan with immediate effect.
- B Ensure that performance is objectively and transparently assessed.
- C Give another assignment to Vardarajan.
- D Give higher salary to Vardarajan.
- E Fire Vardarajan.

32. Over the last five years, Bank of Bharat has seen the number of its retail customer accounts drop by over 40 percent. Over the same period, the share price of Bank of Bharat has increased by more than 80 percent. This amazed a few investors, who believe that a bank's share price should drop if its number of customers drops. Which of the following, if true over the last five years, best accounts for the observed movement in the price of Bank of Bharat's equity shares?

- A Two years ago Securities and Exchange Board started an investigation on the bank for accounting irregularities, but last year the company was cleared of all charges.

- B The bank recently implemented a highly publicized program for free home loans.
- C The bank has been switching its customer base from retail customers to commercial customers, which now accounts for over 75 percent of the bank's revenues.
- D There have been many new banks, which have entered retail banking business over the last five years.
- E The bank is known to be the best paymaster in the industry.



33. Gastric bypass surgery has been shown to be effective at helping extremely obese people lose weight. Some patients have lost as much as 300 pounds after undergoing the surgery, thereby substantially prolonging their lives. Despite the success of the treatment, most doctors have not embraced the surgery. Which of the following statements, if true, best accounts for the lukewarm reaction of the medical community to gastric bypass surgery?
- A Gastric bypass surgery carries a high risk of serious complications, including death.
 - B Obesity is one of the leading contributors to heart disease and hypertension, two leading causes of death.
 - C Incidences of obesity among the Indian urban middle class population have been increasing consistently for the last three decades.
 - D Many patients report that losing weight through diets is ineffective, since they usually gain the weight back within six months.
 - E Most health insurance plans will cover the cost of gastric bypass surgery for morbidly obese patients at high risk of heart disease.

Instructions [34 - 35]

Read the following caselet and choose the best alternative:

Island of Growth was witnessing a rapid increase in GDP. Its citizens had become wealthier in recent times, and there had been a considerable improvement in the standards of living. However, this rapid growth had increased corruption and nepotism in the Island. In the recent times, a fear had gripped the population that corruption would destroy the inclusive nature of the society and hinder economic progress. However, most citizens had kept quiet because:

- a. they had benefited from the corruption indirectly, if not directly.
- b. they did not have the time and energy to protest.
- c. they did not have courage to rise against the established power centers.

There was a need to remove corruption but no one was willing to stick his neck out. Many politicians, bureaucrats and private organizations were corrupt. Media and intellectuals kept quiet, as they benefited indirectly from corruption. The common man was scared of state's retribution and the youngsters feared insecure future.

Against this background, an old, unmarried and illiterate gentleman of high moral and ethical authority, Shambhu, decided to take on the issue of corruption. He sat on a hunger strike in the heart of the capital city of the Island. Shambhu demanded that the Government should constitute new laws to punish the corrupt across all walks of life. Media and the citizens of the island gave massive support to Shambhu. Buckling under the pressure, the Government promised to accept Shambhu's demands. He ended the hunger strike immediately following the Government's announcement. Shambhu became the darling of the media. He used this opportunity as a platform to spread the message that only citizens with an unblemished character should be

allowed to hold a public office.

A few months later, it was found that the Government had not fulfilled any of its promises made to Shambhu. Infuriated, he was thinking of launching another island-wide protest. However, this time, he sensed that not many people and media persons were willing to support him.

34. Read the following statements:

1. People's latent anger against corruption
2. Shambhu's moral courage
3. Hungry media looking to raise issues
4. Raising income level

In your opinion, which combination of the above statements is the most unlikely reason for Shambhu's initial success?

- A 1,2,4
- B 3,4
- C 1,3
- D 1,3,4
- E 1,2,3,4

35. Which of the following could be the most likely reason for decline in public support for Shambhu?

- A The common man had become sick and tired of Government's inaction against rising corruption.
- B Shambhu was old and he lacked the energy to garner the same support that he enjoyed from the media and the public in the initial stages.

The general public may have realized that Shambhu was focusing too much on "indirect involvement" in past incidences of corruption. Common men found it difficult to live up to the high standards set by Shambu.
- C
- D Shambhu's colleagues were misleading him.
- E Shambhu came from a village, while most of his supporters were city dwellers.



Instructions [36 - 38]

Read the following caselet and choose the best alternative:

Head of a nation in the Nordic region was struggling with the slowing economy on one hand and restless citizens on the other. In addition, his opponents were doing everything possible to discredit his government. As a famous saying goes, "There is no smoke without a fire", it cannot be said that the incumbent government was doing all the right things. There were reports of acts of omission and commission coming out every other day.

Distribution of public resources for private businesses and for private consumption had created a lot of problems for the government. It was being alleged that the government has given the right to exploit these public resources at throw-away prices to some private companies. Some of the citizens were questioning the government policies in the Supreme Court of the country as well as in the media. In the midst of all this, the

He asked his minister of water resources about the bidding process for allocation of rights to setup mini-hydel power plants. To this, the minister replied that his ministry had followed the laid out policies of the government. Water resources were allocated to those private companies that bid the highest and were technically competent. The minister continued that later on some new companies had shown interest and they were allowed to enter the sector as per the guidelines of the Government. This, the minister added, would facilitate proper utilization of water resources and provide better services to the citizens. The new companies were allocated the rights at the price set by the highest bidders in the previous round of bidding. After hearing this, the head of the nation replied that one would expect the later allocations to be done after a fresh round of bidding. The minister of water resources replied that his ministry had taken permissions from the concerned ministries before allocating the resources to the new companies.

- A** If benefit to the citizens is higher than notional losses then it is not unethical.
- B** If benefit to the citizens is lower than notional losses then it is unethical.
- C** If benefit to the citizens is higher than actual losses then it is not unethical.
- D** If benefit to the citizens is lower than actual losses then it is unethical.
- E** All of the above A, B, C and D are inappropriate.

- A** Media has been rightly accused by the minister.
- B** Minister's statement may be factually incorrect.
- C** Media has rightly accused the minister.
- D** Media has wrongly accused the minister.
- E** None of above

- A** Only corporate managers.
- B** Only the minister.
- C** Only the Head of the nation.
- D** All of the above.
- E** None of the above.

Instructions [39 - 42]

Read the following caselet and choose the best alternative

The BIG and Colourful Company

You are running "BIG and Colourful (BnC)" company that sells books to customers through three retail formats:

- You can buy books from bookstores,
- You can buy books from supermarket,
- You can order books over the Internet (Online).

Your manager has an interesting way of classifying expenses: some of the expenses are classified in terms of size: Big, Small and Medium; and others are classified in terms of the colors, Red, Yellow, Green and Violet. The company has a history of categorizing overall costs into initial costs and additional costs. Additional costs are equal to the sum of Big, Small and Medium expenses. There are two types of margins, contribution (sales minus initial costs) and profit (contribution minus additional costs). Given below is the data about sales and costs of BnC:

Sales		60000
Initial Costs		39000
Contribution (Sales - Initial Costs)		21000
Additional Costs		
Big	9300	
Small	3000	
Medium	3500	
		15800
Profit (Contribution - Additional Cost)		5200

Each of the Big, Small and Medium cost is categorized by the manager into Red, Yellow, Green and Violet costs. Breakdown of the additional costs under these headings is shown in the table below:

Expenses	Total	Red	Yellow	Green	Violet
Big	9300	5100	1200	1400	1600
Small	3000		400	2000	600
Medium	3500	400	1500	1400	200
Total	15800	5500	3100	4800	2400

Red, Yellow, Green and Violet costs are allocated to different retail formats. These costs are apportioned in the ratio of number of units consumed by each retail format. The number of units consumed by each retail format is given in the table below:

Retail Format	Red	Yellow	Green	Violet
Online	200	52	50	50
Supermarket	65	20	21	21
Book Store	10	30	9	9
Total	275	100	80	80

39. Read the following statements:

Statement I. Online store accounted for 50% of the sales at BnC and the ratio of supermarket sales and book store sales is 1:2.

Statement II. Initial Cost is allocated in the ratio of sales. If you want to calculate the profit/loss from the different retail formats, then

- A Statement I alone is sufficient to calculate the profit/loss.
- B Statement II alone is sufficient to calculate the profit/loss.
- C Both statements I and II are required to calculate the profit/loss.
- D Either of the two statements is sufficient to calculate the profit/loss.
- E Neither Statement I nor Statement II is sufficient to calculate the profit/loss.

40. What is the profit/loss from "online" sales? (Use data from previous question if needed)

- A 0
- B -310
- C +20
- D +388
- E Cannot be determined from given information

41. Which retail format is least profit making for BnC?

- A Online
- B Supermarket
- C Book Store
- D All formats are loss making. E. All formats are profit making.



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42. Which retail format gives the highest profit for BnC?

- A BookStore
- B Supermarket
- C Online
- D All are equally profitable.
- E Cannot be determined from given information.

Instructions [43 - 44]

Read the following caselet and choose the best alternative

A teacher wanted to administer a multiple choice (each question having six choices) based quiz of high difficulty level to a class of sixty students. The quiz had sixty questions. The probability of selecting the correct answer for a good student and a brilliant student was 0.2 and 0.25 respectively. The poor students had no learning advantage. The teacher did not want students to cheat but does not have time and resources to monitor. All students were seated serially in 10 rows and 6 columns.

43. Is it possible for the teacher to detect cheating without monitoring? Choose the statement that best describes your opinion:

- A It is not at all possible; teacher will have to introduce technology if there is no human support.
- B It is always possible; but teacher has to calculate the exact answer.
- C It is possible when many students sitting next to each other have the same incorrect answers for multiple questions. However, there can be a small error in judgment.
- D It is possible when many students sitting next to each other have the same correct answers for multiple questions. However, there can be a small error in judgment.
- E It is possible only for poor students but not for good and brilliant students. However, there can be a small error in judgment.

44. Three good students were seated next to each other. What is the probability of them having the same incorrect choice for four consecutive questions?

- A $256/390625$
- B $256/3125$
- C $4/3125$
- D $1/3125$
- E Cannot be calculated



45. Students from four sections of a class accompanied by respective class teachers planned to go for a field trip. There were nineteen people in all. However, on the scheduled day one of the four teachers and a few students could not join the rest. Given below are some statements about the group of people who ultimately left for the trip.

- I: Section A had the largest contingent.
- II: Section B had fewer students than Section A.
- III: Section C's contingent was smaller than Section B.
- IV: Section D had the smallest contingent.
- V: The product of the number of student from each section is a multiple of 10.
- VI: The number of students from Section C is more than 2.
- VII: The product of the number of students from each section is a multiple of 24.
- VIII: The largest contingent has more than 4 students.
- IX: Each section contributed different number of students

The statements that taken together can give us the exact number of students from each section:

- A I, II, III, IV, VI
- B I, VI, VIII, IX
- C I, II, III, IV, V, VI
- D I, II, III, IV, VI, VII
- E I, IV, VI, VII, IX

46. There is a lot of interest in the first five ranks for Class XI students. One student guessed the rank order as Ankita, Bhagyashree, Chanchal, Devroopa and Esha. Later upon announcement of the results, it was found that not only did he get each student out of her true position, none of the students in his ranking correctly followed her immediate predecessor. Another student guessed Devroopa, Ankita, Esha, Chanchal and Bhagyashree. Even his guess was wrong. It was found that he had got two positions correct, and two students in his ranking correctly followed their immediate predecessors. Which of the following is true about the correct rank order?

- A Ankita got the third position
- B Bhagyashree got the fourth position.
- C Chanchal got the second position.
- D Devroopa stood first.
- E Esha got the fourth position.

Instructions [47 - 50]

Read the following caselet and choose the best alternative:

Prof. Vijya, the chairperson of the Faculty Academic Committee (FAC), was trying to understand the implications of decisions taken by the Student Placement Committee (SPC) on placement issues.

It was alleged that Biswajit, a final year student, inflated his grades in his bio-data that was sent to the recruiters. The President of SPC requested the FAC to debar Biswajit from the campus recruitment process. When the matter was brought up for discussion in FAC, one of the professors remarked that Biswajit too should be allowed to defend himself. When Biswajit arrived for the meeting the situation became even more challenging.

Biswajit raised the issue that many other students who had misrepresented grades to get coveted jobs had gone scot-free. He alleged that these students were close to the President of SPC and therefore, no action was taken against them. He stated that somebody has deliberately manipulated his grades in the bio-data.

This allegation confused the members and it was decided to adjourn the meeting. Vijya was to decide on the next course of action.

47. If you were Vijya what in your opinion would be the most appropriate action?

- A Debar Biswajit and the President of SPC from the placement process as they have failed to uphold the rules of the SPC.
- B Suspend Biswajit and the President of SPC.
- C Constitute a fact finding committee to investigate the matter and ask them to submit a report to you within a week's time.
- D Apprise the corporate recruiters of the situation and assure them that corrective actions will be taken.
- E Both options C and D.



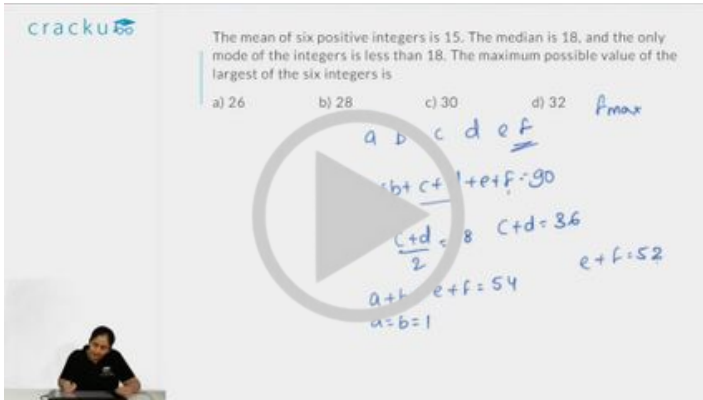
48. It was found that a large section of the students have been indulging in such practices. Unfortunately, the HR manager of a much coveted campus recruiter, who is an alumnus of the college came to know about this. Considering yourself in the position of that HR manager, what would be your reaction?
- A Express your displeasure and stop any further recruitment from the college.
 - B Talk to your contact in the college and try to find out the truth.
 - C Do not change anything and continue the process as if nothing has happened.
 - D Ask the college to send a fresh set of bio-data as you wanted verified grades of the students.
 - E Ask the SPC to resend the details of the shortlisted students including their verified grades.
49. As a potential entrant you are having an informal Facebook chat with one of the college seniors. You wanted to know about the pay packages of the graduating students. The senior replied that one will be able to understand this only after joining the college. He did not reveal any information but suggested that it is not very difficult to get a high salary job as you have already started networking. He also cautioned that you should not believe any rumors and you should directly contact the student body for any further information. What will be your most appropriate choice for seeking further information about placements?
- A This conversation will increase your interest and you will network more with the students of the college for increasing your chances of getting a high salary job.
 - B You will contact the college authorities to get more insights about the placement process.
 - C You will start networking with the HR managers to understand their requirements.
 - D You will try to contact the President of SPC
 - E You will contact your other Facebook friends to find out about the placement activities at the college.
50. Vijya found that there were many such cases of grade inflation. She was giving final touches to the report when her attention was diverted by a phone call on her personal cell phone. It was from an unlisted number. The caller conveyed to her that it will be in her interest as well as in the interest of the college if the report is not presented to the Director. The caller also told her these findings will change nothing only result in bad publicity for the college. The caller identified himself as a well-wisher of the college before hanging up. Consider yourself in Vijya's position and choose an appropriate decision from the following choices.
- A Disregard the phone call and do not share its details with others.
 - B Understand the implications of the phone call and apply for a leave.
 - C Call up the Director, tell him about the phone call, and excuse yourself from the responsibility.
 - D Talk to the Director and seek his opinion.
 - E Constitute a different committee to investigate the "phone call" and carry on with normal Activities.



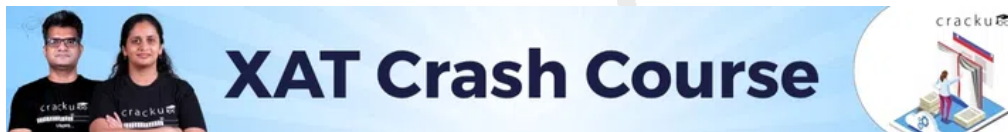
Quantitative Ability and DI

51. The mean of six positive integers is 15. The median is 18, and the only mode of the integers is less than 18. The maximum possible value of the largest of the six integers is

- A 26
- B 28
- C 30
- D 32
- E 34



VIDEO SOLUTION



52. Ramesh bought a total of 6 fruits (apples and oranges) from the market. He found that he required one orange less to extract the same quantity of juice as extracted from apples. If Ramesh had used the same number of apples and oranges to make the blend, then which of the following correctly represents the percentage of apple juice in the blend?
- A 25%
 - B 33.3%
 - C 60%
 - D 66.6%
 - E None of the above.
53. Sara has just joined Facebook. She has 5 friends. Each of her five friends has twenty five friends. It is found that at least two of Sara's friends are connected with each other. On her birthday, Sara decides to invite her friends and the friends of her friends. How many people did she invite for her birthday party?
- A ≥ 105
 - B ≤ 123
 - C < 125
 - D ≥ 100 and ≤ 125

E ≥ 105 and ≤ 123

54. Prof. Mandal walks to the market and comes back in an auto. It takes him 90 minutes to make the round trip. If he takes an auto both ways it takes him 30 minutes. On a Sunday, he decides to walk both ways. How long would it take him?

- A 100 minutes
- B 120 minutes
- C 140 minutes
- D 150 minutes
- E None of the above

[VIDEO SOLUTION](#)

55. At the centre of a city's municipal park there is a large circular pool. A fish is released in the water at the edge of the pool. The fish swims north for 300 feet before it hits the edge of the pool. It then turns east and swims for 400 feet before hitting the edge again. What is the area of the pool?

- A 62500π
- B 125000π
- C 250000π
- D 500000π
- E Cannot be answered from the given data.

56. Mr. Mehra is planning for higher education expenses of his two sons aged 15 and 12. He plans to divide Rs 15 lakhs in two equal parts and invest in two different plans such that his sons may have access to Rs 21 lakhs each when they reach the age of 21. He is looking for plans that will give him a simple interest per annum. The rates of interest of the plans for his younger son and his elder son should be

- A 5% and 7.5% respectively

58. The taxis plying in Wasseypur have the following fare structure: Rs 20 for the first two kilometers, Rs 5 for every km in excess of 2 km and up to 10 km, and Rs 8 for every km in excess of 10 km. Bullock carts on the other hand charge Rs 2 per km. Sardar Khan takes a taxi from the Wasseypur railway station to his home. On the way, at a distance of 14 km from the railway station, he meets Faizal Khan, and gets down from the taxi to talk to him. Later he takes a bullock cart to reach his home. He spends a total of Rs 102 to reach his home from the railway station. How far is his home from the railway station (in kilometers)?
- A 17
B 18
C 19
D 20
E 21
59. Consider the expression $\frac{(a^2+a+1)(b^2+b+1)(c^2+c+1)(d^2+d+1)(e^2+e+1)}{abcde}$, where a,b,c,d and e are positive numbers. The minimum value of the expression is
- A 3
B 1
C 10
D 100
E 243

Consider the expression $\frac{(a^2+a+1)(b^2+b+1)(c^2+c+1)(d^2+d+1)(e^2+e+1)}{abcde}$, where a,b,c,d and e are positive numbers. The minimum value of the expression is

A) 3
B) 1
C) 10
D) 100
E) 243



$$\left(a + 1 + \frac{1}{a}\right) \left(b + 1 + \frac{1}{b}\right) \left(c + 1 + \frac{1}{c}\right) \left(d + 1 + \frac{1}{d}\right) \left(e + 1 + \frac{1}{e}\right)$$



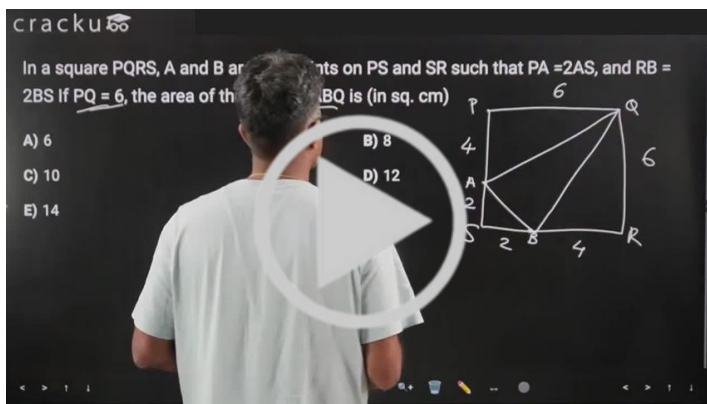


 VIDEO SOLUTION

60. In a square PQRS, A and B are two points on PS and SR such that PA = 2AS, and RB = 2BS. If PQ = 6, the area of the triangle ABQ is (in sq. cm)
- A 6
B 8
C 10

D 12

E 14



VIDEO SOLUTION



61. How many whole numbers between 100 and 800 contain the digit 2?

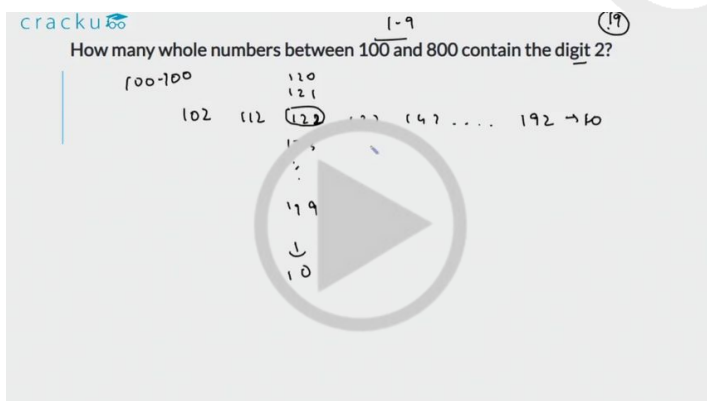
A 200

B 214

C 220

D 240

E 248



VIDEO SOLUTION

62. p , q and r are three non-negative integers such that $p + q + r = 10$. The maximum value of $pq + qr + pr + pqr$ is

A ≥ 40 and < 50

B ≥ 50 and < 60

C ≥ 60 and < 70

D ≥ 70 and < 80

E ≥ 80 and < 90

p, q and r are three non-negative integers such that $p + q + r = 10$. The maximum value of $pq + qr + pr$ is

A) ≥ 40 and < 50
B) ≥ 50 and < 60
C) ≥ 60 and < 70
D) ≥ 70 and < 80
E) ≥ 80 and < 90

Handwritten notes: $p(2q+2r) = \frac{10 \times 10}{3}$, $2q+2r+1 = 10$



VIDEO SOLUTION

63. A number is interesting if on adding the sum of the digits of the number and the product of the digits of the number, the result is equal to the number. What fraction of numbers between 10 and 100 (both 10 and 100 included) is interesting?

A 0.1

B 0.11

C 0.16

D 0.22

E None of the above

cracku

A number is interesting if on adding the sum of the digits of the number and the product of the digits of the number, the result is equal to the number. What fraction of numbers between 10 and 100 (both 10 and 100 included) is interesting?

A) 0.1
B) 0.11
C) 0.16
D) 0.22
E) None of the above

Handwritten notes: $100 \Rightarrow 1+0+0 + 1 \times 0 \times 0$, $0, 1$ to $9, 0$, $0+0=0$ or $0+0=0$



VIDEO SOLUTION

cracku

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64. 70% of the students who joined XCRI last year play football, 75% play cricket, 80% play basketball and 85% play carrom. The minimum percentage of students who play all four games is:
- A 5%
- B 10%
- C 15%
- D 20%
- E None of the above
65. p and q are positive numbers such that $p^q = q^p$, and $q = 9p$. The value of p is
- A $\sqrt{9}$
- B $\sqrt[6]{9}$
- C $\sqrt[9]{9}$
- D $\sqrt[8]{9}$
- E $\sqrt[3]{9}$
66. Ram, Shyam and Hari went out for a 100 km journey. Ram and Hari started the journey in Ram's car at the rate of 25 kmph, while Shyam walked at 5 kmph. After sometime, Hari got off and started walking at the rate of 5 kmph and Ram went back to pick up Shyam. All three reached the destination simultaneously. The number of hours required for the trip was:
- A 8
- B 7
- C 6
- D 5
- E 4

cracku

Ram, Shyam and Hari went out for a 100 km journey. Ram and Hari started the journey in Ram's car at the rate of 25 kmph, while Shyam walked at 5 kmph. After sometime, Hari got off and started walking at the rate of 5 kmph and Ram went back to pick up Shyam. All three reached the destination simultaneously. The number of hours required for the trip was:

A) 8
B) 7
C) 6
D) 5
E) 4

Handwritten solution:

Let T_1 be the time taken for the first part of the journey. The distance covered by the car is $25T_1$. The distance covered by Shyam is $5T_1$. The distance covered by Hari is $100 - 25T_1$. The time taken for Hari to reach the destination is $T_1 + \frac{100 - 25T_1}{5}$. The time taken for the car to return to pick up Shyam is $\frac{25T_1}{5}$. The total time taken for the car to return to pick up Shyam is $T_1 + \frac{25T_1}{5}$. The total time taken for the car to return to pick up Shyam is $T_1 + \frac{25T_1}{5}$.

Diagram showing the journey of Ram, Shyam, and Hari. The car travels a distance of $25T_1$ and then returns to pick up Shyam. Shyam walks a distance of $5T_1$. Hari walks a distance of $100 - 25T_1$.

VIDEO SOLUTION



67. The central park of the city is 40 metres long and 30 metres wide. The mayor wants to construct two roads of equal width in the park such that the roads intersect each other at right angles and the center of the rectangle co-incides with the center of the rectangle formed by the intersection of the two roads . Further, the mayor wants that the area of the two roads to be equal to the remaining area of the park. What should be the width of the roads?

- A** 10 metres
- B** 12.5 metres
- C** 14 metres
- D** 15 metres
- E** 16 metres

68. Arun has to go to the country of Ten to work on a series of tasks for which he must get a permit from the Government of Ten. Once the permit is issued, Arun can enter the country within ten days of the date of issuance of the permit. Once Arun enters Ten, he can stay for a maximum of ten days. Each of the tasks has a priority, and takes a certain number of days to complete. Arun cannot work on more than one task at a time. The following table gives the details of the priority and the number of days required for each task.

Task	Priority	Number of Days Required
T1	1	3
T2	2	5
T3	5	3
T4	3	4
T5	4	2

Arun's first priority is to complete as many tasks as possible, and then try to complete the higher priority tasks. His last priority is to go back as soon as possible.

The tasks that Arun should try to complete are:

- A** T1 and T2
- B** T1, T2 and T5
- C** T1, T4 and T5
- D** T1, T2 and T4
- E** T1, T3 and T4

69. Arun has to go to the country of Ten to work on a series of tasks for which he must get a permit from the Government of Ten. Once the permit is issued, Arun can enter the country within ten days of the date of issuance of the permit. Once Arun enters Ten, he can stay for a maximum of ten days. Each of the tasks has a priority, and takes a certain number of days to complete. Arun cannot work on more than one task at a time.

However, Arun's manager has told him to do some background research on the tasks before leaving for Ten. At the same time, there is no guarantee that the Government of Ten will give the permit to Arun. Background research involves substantial costs, and therefore Arun has decided that he will not start his background research without getting the permit.

The following table gives the details of the priority, the number of days required for each task and the number of days required for background research on each task.

Task	Priority	Number of Days Required	No. of Days Required for Background Research
T1	1	3	3
T2	2	5	5
T3	5	3	2
T4	3	4	2
T5	4	2	3

Arun's first priority is to complete as many tasks as possible, and then try to complete the higher priority tasks. His last priority is to go back as soon as possible within ten days.

The tasks that Arun should try to complete are:

- A T1, T2 and T3
- B T1, T2 and T5
- C T1, T2 and T4
- D T1, T3 and T4
- E T1, T4 and T5



70. The radius of a circle with centre O is $\sqrt{50}$ cm. A and C are two points on the circle, and B is a point inside the circle. The length of AB is 6 cm, and the length of BC is 2 cm. The angle ABC is a right angle. Find the square of the distance OB.

- A 26
- B 25
- C 24
- D 23
- E 22

71. Six playing cards are lying face down on a table, two of them are kings. Two cards are drawn at random. Let a denote the probability that at least one of the cards drawn is a king, and b denote the probability of not drawing a king. The ratio a/b is

- A ≥ 0.25 and < 0.5
- B ≥ 0.5 and < 0.75
- C ≥ 0.75 and < 1.0
- D ≥ 1.0 and < 1.25
- E ≥ 1.25

cracku

Six playing cards are lying face down on a table, two of them are kings. Two cards are drawn at random. Let a denote the probability that at least one of the cards drawn is a king, and b denote the probability of not drawing a king. The ratio a/b is

A) ≥ 0.25 and < 0.5
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 C) ≥ 0.75 and < 1.0
 D) ≥ 1.0 and < 1.25
 E) ≥ 1.25

Handwritten notes: $6C_2 = 15$, $4 = 6$

Hand-drawn diagram showing two kings (K) and four non-kings (represented by squares) with a play button icon in the center.

VIDEO SOLUTION

72. In the country of Four, there are four cities, A, B, C and D. B is to the East of A, C is to the South of B, D is to the West of C, and A is to the North of D. The Government of Four is planning to connect these four cities by road such that it is possible for a person to go from a city to any of the other three cities. At the same time, the Government wants to ensure that the total road length is minimum. The distances between A to B, B to C, C to D and D to A are all equal to 10 km. What should be the total length of the road?

- A 26.64 km
- B 27.32 km
- C 28.30 km
- D 30 km
- E 36 km

cracku

XAT Score Calculator

73. Consider the expression $(xxx)_b = x^3$, where b is the base, and x is any digit of base b . Find the value of b :

- A 5
- B 6
- C 7
- D 8
- E None of the above

74. Consider a function $f(x) = x^4 + x^3 + x^2 + x + 1$, where x is a positive integer greater than 1. What will be the remainder if $f(x^5)$ is divided by $f(x)$?

- A 5
- B 6
- C 7

D a monomial in x

E a polynomial in x

75. Please read the following sentences carefully:

I – 103 and 7 are the only prime factors of 100027

II – $\sqrt[6]{6!} > \sqrt[7]{7!}$

III – If I travel one half of my journey at an average speed of x km/h, it will be impossible for me to attain an average speed of $2x$ km/h for the entire journey.

A All the statements are correct

B Only Statement II is correct

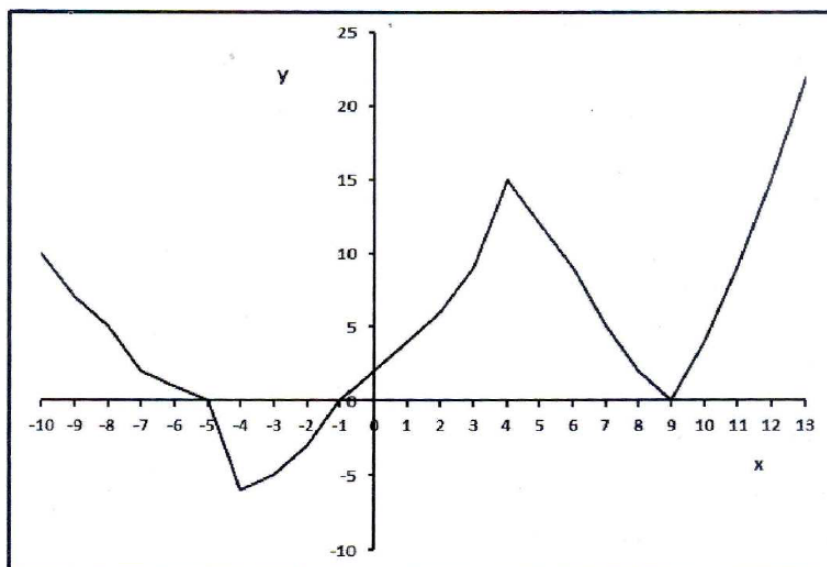
C Only Statement III is correct

D Both statements I and II are correct

E Both statements I and III are correct



76. The figure below shows the graph of a function $f(x)$. How many solutions does the equation $f(f(x)) = 15$ have?



A 5

B 6

C 7

D 8

E cannot be determined from the given graph

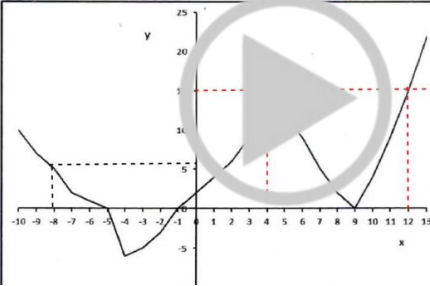
The figure below shows the graph of a function $f(x)$. How many solutions does the equation $f(f(x)) = 15$ have?

$$f(-8) = 6$$

$$f(f(x)) = 15$$

$$f(a) = 15$$

$$f(4) = 15$$



VIDEO SOLUTION

Instructions [77 - 79]

Given below are a few data points on the Indian economy from 2005 to 2010:

Indicator	Unit	2005	2006	2007	2008	2009	2010
GDP, current prices	Rs Billions	35662.2	41159.73	47675.86	54470.27	60712.76	73555.34
GDP, per capita current prices	Rs	32128.1	36553.93	41747.69	47038.23	51714.45	61784
Gross national savings Percent of GDP	%	32.88	34.28	36.65	32.17	35.08	32.14
Inflation, average consumer prices index	Index	115.67	122.92	130.75	141.67	157.08	175.92
Volume of imports of goods and services	% Change	17.99	9.438	16.3	10.84	8.321	16.49
Volume of exports of goods and services	% Change	18.88	13.83	17.13	10.63	0.813	21.86
Unemployment rate	%	9.2	8.9	7.8	7.2	6.8	7.32
Current account balance percent of GDP	%	-1.272	-1.024	-0.701	-2.475	-2.066	-3.268

*per capita GDP is arrived by dividing GDP by population.

77. What is the ratio of the current account balance in 2010 to the current account balance in 2005?

- A 0.35
- B 4.56
- C 5.01
- D 2.57
- E 5.30

78. Read the statements given below:

1. Exports were more than imports in 2006
 2. Imports were more than exports in 2009
 3. Exports increased at faster rate than imports during the period 2005 to 2010
- Which of the above statements is necessarily true?

- A 1 and 2
- B 1, 2 and 3
- C 3 only
- D 2 only
- E 1 only



XAT Crash Course

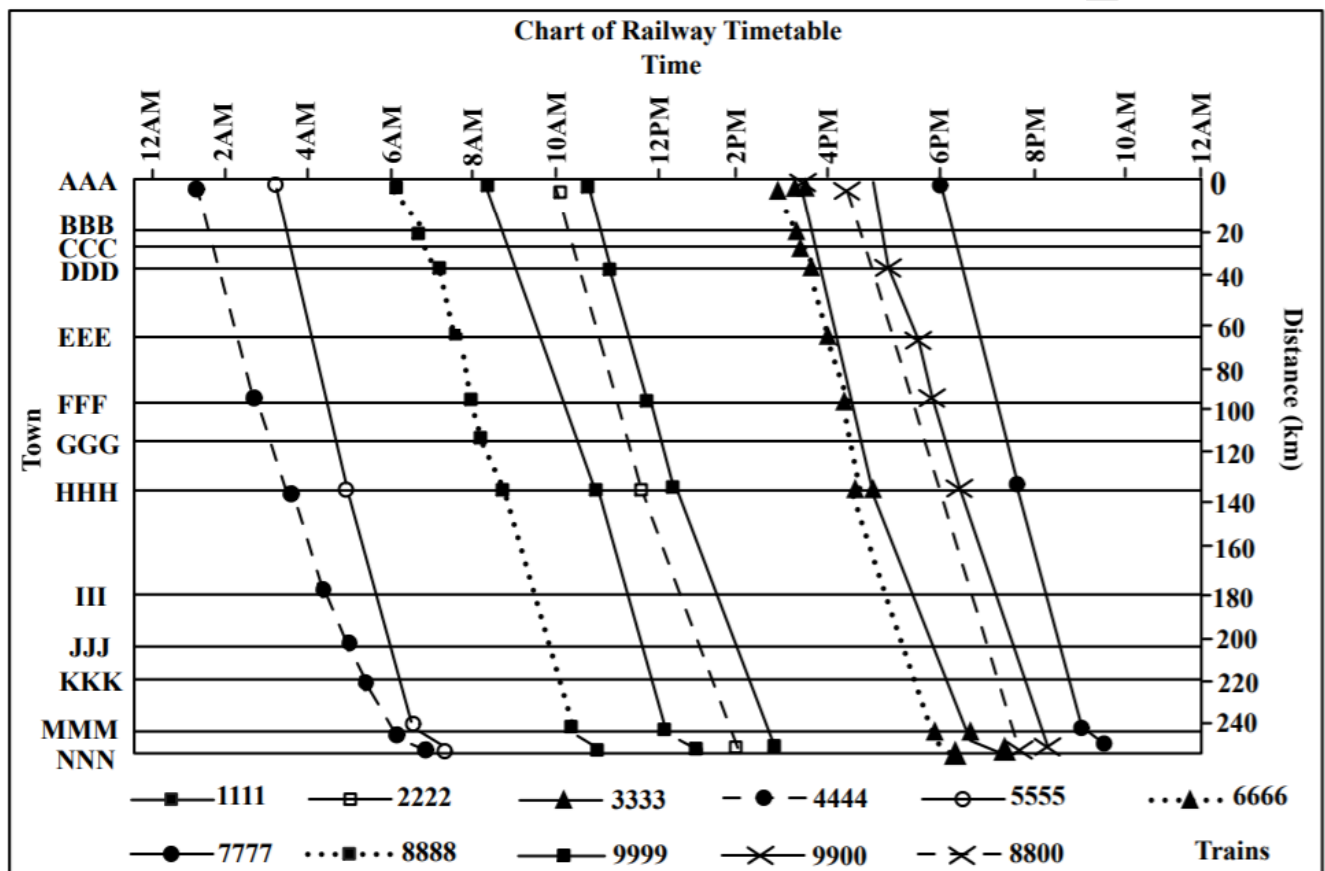


79. What was the approximate number of unemployed persons in 2006?

- A 100 million
- B 102 million
- C 98 million
- D 105 million
- E 104 million

Instructions [80 - 82]

Analyze the railway train chart below:



80. Which is the fastest train between HHH and NNN?

- A 1111
- B 2222
- C 4444
- D 7777
- E 9999

81. Which is the fastest train between AAA and NNN?

- A 8800
- B 8888
- C 6666
- D 4444
- E 3333



82. If you have to travel from AAA and reach HHH at around 9:00 AM, and then further travel to NNN at around 6:00 PM, which is the best combination of trains for you?

- A 1111 and 8800
- B 1111 and 9900
- C 8888 and 7777
- D 8888 and 8800
- E 8888 and 9900

Instructions [83 - 86]

Answer questions based on the following information:

Data on an ongoing football league of a country is given below. 20 teams are playing in the league. The rules of the league are as follows:

1. Each team plays all the other teams twice, once in its home ground, and once in the opponent's home ground. These matches are known as the "Home" match and the "Away" match respectively.
2. A win results in 3 points, a draw in 1 point, and a loss in 0 point for the team.
3. The number of goals a team scores is termed as "Goals For" and the number of goals it concedes is termed as "Goals Against". We get the "Goal Difference" by subtracting "Goals Against" from "Goals For".

The ranking of the teams is decided on the total points. If two teams are tied on their total points, the team which has a higher Goal Difference gets the higher rank. If the tie cannot be resolved on Goal Difference, Goals For is checked followed by Goals Against. If the tie persists, the teams are ranked in the ascending order of their names.

Table 1 provides data on the current top 13 teams based on the overall situation, i.e., by taking into account both home matches and away matches of each team.

Table 2 provides data on the current top 13 teams based on home matches only.

Chart 1 provides a plot of the goal difference of each of the 13 teams based on the overall situation.

Overall				
Team	M	W	D	GF
CH	3	3	0	8
SW	3	2	1	10
WB	3	2	1	6
MC	3	2	1	8
MU	3	2	0	6
WH	3	2	0	4
EV	3	2	0	4
AS	3	1	2	2
WG	3	1	1	4
NC	3	1	1	3
FU	3	1	0	7
ST	3	0	3	3
SU	2	0	2	2

Table 1

Home					
Team	M	W	D	GF	GA
WB	2	2	0	5	0
WH	2	2	0	4	0
CH	2	2	0	6	2
MC	2	2	0	6	3
SW	2	1	1	5	2
NC	2	1	1	3	2
FU	1	1	0	5	0
EV	1	1	0	1	0
MU	1	1	0	3	2
TOT	2	0	2	2	2
NW	1	0	1	1	1
AS	1	0	1	0	0
ST	1	0	1	0	0

Table 2

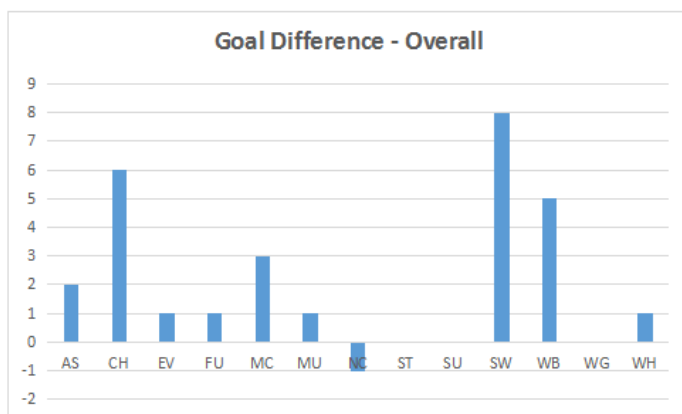


Chart 1

83. Considering away matches only, which of the following teams is the second ranking team?

- A AS
- B CH
- C WG
- D SW
- E WB

84. Considering away matches only, the least number of teams with either 0 or 1 point is:

- A 2
- B 3
- C 4
- D 5
- E 6

XAT Important Topics

85. Let us define a term pos as the difference between "home rank" and "away rank". Which of the following has the maximum value of pos?

- A AS
- B WB
- C WH
- D MC
- E SW

86. How many unique values of goal difference are there for away matches?

- A 5
- B 6
- C ≥ 6
- D ≥ 6 and ≤ 13
- E ≥ 6 and ≤ 14



General Awareness

87. Which of the following Indian companies feature in Fortune Five hundred list for 2012?

- A Reliance Industries, Tata Motors, Tata Steel, IOC
- B Tata Motors, SBI, CTS, Wipro, IOC
- C Tata Steel, IOC, Reliance Industries, Ranbaxy
- D IOC, Reliance Industries, SBI, GAIL
- E IOC, Reliance Industries, ACC, BPCL



88. Which of the following bodies decide the monetary policy in India?

- A Reserve Bank of India
- B Ministry of Finance

- C Ministry of Commerce
- D Ministry of Finance and Commerce
- E Planning Commission

89. CPI and WPI as acronyms refer to

- A Political Parties
- B Purchasing capacity indices
- C Price Indices
- D Poverty indices
- E Unemployment indices

90. What is Davos famous for?

- A It is best known tourist place of Switzerland
- B G8 members meet here, every year
- C EU financial headquarters
- D World Economic Forum
- E Best Ski resorts in the world



XAT Syllabus PDF



91. Which of the following are millennium development goals?

- A Poverty eradication, reducing child mortality, reducing trade barriers, combating AIDS
- B Poverty Reduction, environment sustainability, universal primary education, equal employment
- C Reduce unemployment, promote diversity, combating malaria, improving maternal health
- D Globalization, reduction in poverty, combating AIDS, universal primary education
- E Poverty eradication, improving maternal health, universal primary education, combating AIDS

92. Which of the countries in Europe were most hit by sovereign debt crisis?

- A Greece, Germany, Spain, Iceland
- B Portugal, Spain, Italy, Czech Republic
- C Greece, Spain, France, Switzerland
- D Spain, Greece, Iceland, Portugal
- E Germany, U.K., France, Greece

93. Which of the following countries would account for maximum petroleum production of the World?

- A Saudi Arabia, Iran, Kazakhstan, U.S., Nigeria
- B Russia, U.S., Iran, China, Saudi Arabia
- C Saudi Arabia, U.S., Venezuela, Iran, Iraq
- D Iran, Russia, Saudi Arabia, Nigeria, Libya
- E Iran, Iraq, Saudi Arabia, Libya, Venezuela



94. Which of the following was not the reason for the recent slide of Indian Rupee?

- A Huge trade Deficit
- B Low Growth and High Inflation
- C Low Capital Inflow
- D Burgeoning current account deficit
- E High employment rate

95. Which of the following combinations belongs to Bretton Woods Institute?

- A IMF, World Bank
- B IMF, WTO, Doha Round
- C World, WTO, IBRD
- D IMF, World Bank, UNESCO
- E None of above

96. Read the statements carefully and answer the questions that follow

1. Earth Summit is held every year
2. Last earth Summit took place in Rio De janeiro
3. Another name for earth summit is United Nation Conference on Environment and Development

Which of the above is the right combination ?

- A 1 and 2
- B 2 and 3
- C 1 and 3
- D 1,2 and 3
- E 2 anly



97. Which of following countries meet more than one fourth of their energy requirement from nuclear Power

- A France, U.S., Russia, Germany
- B France, Ukraine, Sweden, Korea
- C France, India, Ukraine, Germany
- D Germany, Russia, Canada, Japan
- E Japan, Germany, Korea, England

98. Pick the right match from the following Combinations

- A Shenzhen - China, NSE - India, SGX Singapore, DAX - Germany
- B NASDAQ - U.S., MCX - India, SGX - Korea, FTSE - France
- C NYSE - U.S., BSE - India, FTSE - England, DAX Denmark
- D NASDAQ - Canada, BSE - India, FTSE France, DAX - Germany
- E Shenzhen - Singapore, Shanghai - China, FTSE - France, BSE - India

99. Harish Hande who won the Magsaysay Award in 2011 is famous for

- A RTE
- B RTI
- C Lokpal
- D MRTP
- E SELCO



100. Indian Government is planning to introduce GAAR. What is the purpose of GAAR?

- A Increase rural Income
- B Increase agriculture income
- C Regulatory authority for managing agriculture resources
- D Regulate foreign investment in agriculture
- E Curtail tax avoidance

101. Who is the chairman of Indian planning Commission?

- A Dr. M.S. Ahluwalia
- B Dr. C. Rangarajan

- C Dr. Manmohan Singh
- D Raminder Singh Gujral
- E Dr. Subba Rao

102. Given below are some well-known awards and their respective fields. Choose the correct combination:

- A Palme d'Or - Television, Pulitzer Advertising, Grammy - Music, Booker Literature
- B Pulitzer - Advertising, Grammy - Music, Cannes Lions - Advertising, Booker Literature
- C Grammy - Music, Books - journalism, Emmy Television, Palme d'Or - Movies
- D Cannes Lions - Movies, Grammy - Music, Pulitzer - journalism, Emmy - Television
- E Palme d'Or - Movies, Pulitzer - journalism, Cannes Lions - Advertising, Booker Literature



103. A few combination names of companies and their products are given below. Choose the right option:

- A Amazon - Kindle, Rovio -Angry Birds, Google - AdSense, Zynga - Nook
- B Amazon - Kindle, Nintendo -Wii, Rovio Angry Birds, Zynga - CastleVille
- C Zynga - Angry Birds, Nintendo -Wii, Google AdSense, Rovio - CastleVille
- D Amazon - Nook, Sony - Wii, Zynga - Castle Ville, Google - Ad sense
- E Amazon - Nook, Nintendo - Wii, Rovio -Castle Ville, Roxio - Kindle

104. Read the following

- I.Excise duty is a tax on manufacture or production of goods and services
 - II.Not all kinds of excise duty is collected by the Central Government of India
 - III.Excise duty is an indirect tax
- Which of the statements given above are correct?

- A Statements I and II
- B Statements II and III
- C Statements I and III
- D All of the above
- E None of the above

105. From which of the following items India export the most?

- A Textile goods
- B lute
- C IT services

- E** Leather



106. Which of the following acids helps in Human Digestion

- E** Acetic

107. Which of the following is the right match for the Olympics medal winners from India?

- E** Saina Nehwal - Uttaranchal, Gaurav Narang Punjab, Sushil Kumar - Haryana, Mary Kom Manipur

108. Which of the following African Countries underwent political turmoil in the last two years?

- E** Egypt, Libya, Lebanon, Algeria, Sudan, Kenya



109. Which of the following education institutes was Mr. Rajat Gupta, the former Mckinsey chief associated with

- B** Indian Institute of Management Calcutta

- C Indian Institute of Management Bangalore
- D Indian Institute of Management Lucknow
- E Indian School of Business Hyderabad

110. Which of country was the highest producer of cars in the world in 2011?

- A U.S.A.
- B Germany
- C Korea
- D Japan
- E China

111. What is the approximate share of manufacturing sector in India GDP in 2011 - 12?

- A 22 %
- B 24 %
- C 31
- D 15 %
- E 10



112. Which of the following combination of scholars have won the Nobel Prize in Economics?

- A Amartya Sen, Joseph Stiglitz, Oliver Williamson
- B Mehboob Haque, Amartya Sen, Robert McNamara
- C Amartya Sen, Mohammad Yunus, Ronald Coase
- D Mohammad Yunus, Paul Krugman, Oliver Williamson
- E Amartya Sen, Elvin Roth, Mohammed Yunus

113. For which of the following reasons was Maruti in the news in July, 2012?

- A Producing most fuel efficient cars
- B Transfer its production to Gujarat
- C Industrial violence
- D Disrupting daily movement of vehicles in New Delhi
- E In July, it became the highest producer of cars in the country

114. What is meaning of one unit carbon permit?

- A One cubic meter of carbon dioxide
- B One quintal carbon dioxide
- C One metric ton of carbon dioxide
- D One thousand metric tons of carbon dioxide
- E One million ounce of carbon dioxide



115. Who is the author of "Imagining India"

- A Khushwant Singh
- B Nandan Nilekani
- C Gurcharan Das
- D Chetan Bhagat
- E APJ Abdul Kalam Azad

116. V Kurien was associated with

- A Operation Flood
- B Operation Milk
- C Operation Blue Star
- D Operation X
- E White Operation



Answers

Verbal and Logical ability

1.A	2.D	3.B	4.E	5.B	6.B	7.B	8.B
9.E	10.B	11.C	12.E	13.A	14.D	15.D	16.E
17.D	18.E	19.C	20.A	21.B	22.D	23.C	24.E
25.D							

Decision Making

26.A	27.E	28.C	29.E	30.A	31.B	32.C	33.A
34.B	35.C	36.E	37.D	38.D	39.C	40.D	41.B
42.A	43.C	44.E	45.E	46.A	47.C	48.D	49.B
50.D							

Quantitative Ability and DI

51.D	52.E	53.B	54.D	55.A	56.E	57.A	58.C
59.E	60.C	61.B	62.C	63.E	64.B	65.D	66.A
67.A	68.B	69.E	70.A	71.E	72.C	73.E	74.A
75.C	76.C	77.E	78.C	79.A	80.D	81.A	82.D
83.D	84.D	85.A	86.B				

General Awareness

87.A	88.A	89.C	90.D	91.E	92.D	93.B	94.E
95.A	96.D	97.B	98.A	99.E	100.E	101.C	102.E
103.B	104.D	105.D	106.C	107.A	108.A	109.E	110.E
111.D	112.A	113.C	114.C	115.B	116.A		

Explanations

Verbal and Logical ability

1. A

'A' should precede the wolf and the lamb since they are introduced for the first time. The 'fold' describes the lamb's fold. Therefore, the article 'the' should be used. Since the wolf and the lamb have already been introduced, the last 2 blanks should be filled with 'the'.

The correct order is a, a, the, the, the. Option A is the right answer.

2. D

'A' should be used to fill the first blank since the bat is introduced for the first time. The article 'the' should be used before 'ground' since the ground is a general term (like the Earth, the Sky). 'A' should be used before the weasel. 'The' should be used before the weasel since the weasel has already been introduced. 'the' should be used before enemy since it states a general concept. 'The' should be used before 'bat' since it has already been introduced. 'A' should be used before both 'bird' and 'rat' since the bat describes himself to be a member of a large group.

A, the, a, the, the, the, a, a is the correct order. Therefore, option D is the right answer.

3. B

The first blank can be filled using 'down' or 'downstairs'. The second blank should be filled using 'as'. 'As usual' is a phrase and hence, 'as' should fill the second blank. The third blank can be filled using 'little' or 'a little'. The 4th blank can be filled with 'out', 'of', or 'out of'. The last blank should be filled by 'to'. Therefore, option B is the right answer.

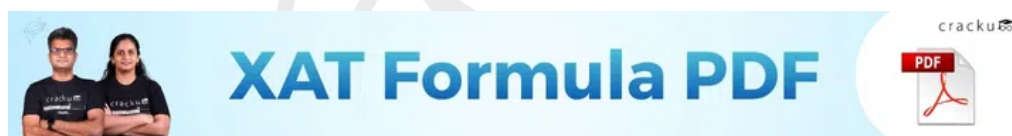
4. E

Statement 1 talks about 'the new laws' being vaguely worded. Statement 2 talks about 'the law' being open to manipulation due to its language. Sentence 1 and 2 are unrelated since 1 talks about a set of new laws (plural) and 2 talks about the law (a particular law that should have been introduced in some other statement, singular). Therefore, the 2 statements are independent and hence, option E is the right answer.

5. B

Statement 1 argues for the free availability of publically financed science. Statement 2 gives information about some possible impediments to free availability. Statement 3 states that the internet revolution is making knowledge transfer cheaper, and in future, it has the possibility of facilitating statement 1.

Thus, the correct option is B.



6. B

The second statement starts with the pronoun 'it'. The third statement talks about the government favouring private sector. Therefore, the 'it' in the second statement most probably refers to the favours extended by the government. Sentences 2 and 3 are dependent. Sentence 1 talks about business schools and is unrelated to the remaining 2 sentences (which talk about the government and politics). Therefore, option B is the right answer.

7. B

The sentence uses both requires and demands, which need not be the case; the structure of the sentence should be requires/demands, and then a list of things that are needed.

This eliminates options A, C and D.

Option E falter at the end with the "but with" introducing a negative condition that does not fit in the sentence's context.

Option B would be the correct replacement, maintaining a parallel structure to the sentence.

8. B

The part to be replaced in the sentence lacks parallel structure and clarity. The phrase "*while standing still*" does not match "*as it does to a person travelling*," making the comparison grammatically inconsistent.

Both parts of the sentence should be the same structurally and grammatically. Option B is the best choice because it ensures a clear and balanced comparison between two people. The phrase "*to a person standing still as to a person traveling*" correctly maintains parallel structure, making the sentence more readable and grammatically sound.

Option A: "*While standing still*" is not parallel to "*as it does to a person traveling*." Here, The phrase "while standing still" lacks context as to who is standing still

Option C: This is not incorrect, but the structure is unnecessarily wordy.

Option D: The phrase "while standing still" suggests a state of being, whereas "to traveling" refers to an action or process. For a proper comparison, both elements should be in the same grammatical form.

Option E: This option is close to being correct but introduces an inconsistency in verb tense. "Standing still" is a continuous action, whereas "who travels" suggests a habitual or general action.

9. E

As savings are in plural form, 'have' should be used. Options A, B and C are eliminated. As the given statement is in passive voice, we use have been instead of have. Therefore, option E is grammatically correct.

The answer is option E.

10. B

Sentences 3 and 4 form a mandatory pair. The author assumes that the man running in the middle is the blind beggar. His assumption gets confirmed on hearing the voice of the beggar.

2 and 1 form a pair. The author first moves out of his position and goes behind the bush. While he is hiding behind the bush, he sees his enemies running on the road. The author thinks that the man running in the middle is the blind beggar and confirms the same on hearing his voice.

2134 is the correct order and hence, option B is the right answer.



11. C

On reading the given sentences, we can infer that the paragraph is about how everyone started running after hearing the pistol sound.

2 must be the opening sentence since it talks about the quarrel that happened before hearing the sound of the horses and the pistol sound.

5 must follow 2 since it starts that 'Almost at the same time' (the time at which they heard the horses).

3 should follow 5 since it states that the pistol sound was the last sign of danger and they started running in all directions deserting Pew.

4 should follow 3 since the pronoun 'him' in the sentence refers to Pew.

1 must be the concluding sentence since it describes how Pew too started running shortly.

The correct order is 25341. Therefore, option C is the right answer.

12. E

1-3 forms a logical pair as the questions mentioned in 3 are given in 1. Thus, options B, C and D are eliminated.

Among 4 and 2, 4 is a better opening statement as it introduces the problem faced by the historians of science.

Hence, the answer is option E.

13. **A**

Fermi determined the lower limit of the yield using the confetti.

"As the wind from the initial blast wave began to blow through the camp, he slowly dribbled the confetti into the air, observing how far back it was scattered by the blast (taking the farthest scattered pieces as being the peak of the pressure wave). Fermi concluded that the yield must be greater than 10 kilotons."

Only option A uses a parallel reasoning. Options B, C, and D try to determine the average and exact values. Therefore, option A is the right answer.

 VIDEO SOLUTION

14. **D**

The last paragraph states that "The value of quick estimates was something Fermi was familiar with throughout his career. He was famous for teaching his students skills at approximation of fanciful-sounding quantities that, at first glance, they might presume they knew nothing about..... But Fermi was trying to teach his students how to solve problems where the ability to confirm the results would not be so easy. He wanted them to figure out that they knew something about the quantity in question."

Fermi emphasized on the value of quick estimates as one can find approximate range of the quantities whose actual quantities are difficult to find.

So Option B,C and E are rejected as finding out the averages or exact values was not the purpose.

Option A is irrelevant.

 VIDEO SOLUTION

15. **D**

The central idea of the passage is that one can estimate any physical quantity, even though the actual value is difficult or not possible to find.

Statement 1 is incorrect as estimation even when the correct value is known is not the central idea of the passage.

Statement 4 incorrect as Fermi's brilliance was not what the author is discussing.

Statement 3 does not contain the central idea of the passage, it is an example for minimal information can provide estimate values.

Hence only statement 2 is correct.

 VIDEO SOLUTION



16. **E**

All the statements cannot be inferred from the passage.

1 "Atomic bomb detonation was a result of Fermi's Nobel Prize contribution" is incoherent with the data. Though both the events occurred in chronological order but the inference is incorrect.

2. "Fermi's students respected him as a scientist" is not mentioned in the paragraph.

3. "Yield of atomic bomb can only be measured in Kilotons " cannot be inferred from the paragraph.

 VIDEO SOLUTION

17. **D**

"In short, we can sometimes deduce that a universal scientific belief is false but we can never induce that a universal scientific belief is true."

Since option D reflects the same idea as the last line of the first paragraph, the correct answer is D.

18. **E**

The idea of 'asymmetry' was extended in the second paragraph to include *"Falsification, that is to say, is no more capable of yielding conclusive rejections of scientific belief than verification is of yielding conclusive acceptances of scientific beliefs."*

Thus, from the above lines of paragraph 2, option E can be inferred.

Thus, the correct option is E.

19. **C**

Option A: This is the opposite of the author's position and hence will not be the correct answer.

Option B: *"Falsifications may not be conclusive, for the acceptances on which rejections are based are always provisional acceptances..."*

From the above lines, it is clear that option B is wrong.

Option C: Although falsifications can sometimes be better at rejecting claims than verifications, in establishing the claims, they are as bad as verification. Since this is in line with the author's position, this is the correct option.

Option D: Although this option is also somewhat in line with the author's position, this will be the second best option. This is because only in some instances falsifications can be better than verification. Since option C aptly captures the author's argument, this option will not be the correct option.

Option E: This option is rejected for the same reason as A.

Thus, the correct option is C.

20. **A**

"...Falsifying claims sometimes give us a good reason for rejecting a scientific belief, namely when the claims are accepted."

From the above lines from the last paragraph, it can be inferred that Falsification gives us a better reason for rejecting a scientific belief.

Thus, the correct option is A.

Option (B) Falsification gives us all the reasons for accepting a claim. (Incorrect)

The passage does not say that falsification supports accepting claims—it only supports rejecting false ones.

Option (C) Verification gives us a reason for rejecting a claim. (Incorrect)

Verification deals with supporting a belief, not rejecting it.

Option (D) Verification gives us an appropriate reason for accepting a scientific belief. (Incorrect)

The passage strongly argues that verification does not give a conclusive reason for accepting a belief.

Option (E) Verification gives us an appropriate reason for rejecting a scientific belief. (Incorrect)

This contradicts the passage, as verification does not involve rejection.



21. **B**

The passage talks about the 'alleged inadequacy of neoclassical economics' when it comes to capitalist thinking. Only option B is congruent with this line of thought, and is the correct answer. Options A and E are contrary to what is stated in the passage while options C and D cannot be inferred from the passage.

22. **D**

The passage says that George Soros thinks the economics used to explain capitalism is too simple and doesn't really show how people actually act. It also mentions that you can't explain everything about humans, like their feelings or beliefs, just by using economic rules. So, basically, the passage argues that the rules of economics can't fully explain all the reasons behind why people do what they do.

The passage clearly states that 'it would be foolish to construe religion or romance or even scientific enquiry as solely explicable by reference to the laws of economics.' Option D is congruent with this point and is the answer.

23. C

According to the passage Soros believes that 'the state can manipulate many aspects of human life', the only rider being that 'it does not settle into any one policy firmly, unbendingly'. Thus it can be inferred that Soros accepted the idea of state intervention provided that such intervention did not ossify into something static. Thus, option C is the answer.

The passage explicitly states that Soros supports state intervention in economic matters.

Soros is not a supporter of laissez-faire capitalism.

Soros distrusts free-market capitalism, as stated in the passage.

24. E

'Critical rationalism' finds mention in the passage as that attribute which 'Soros himself regarded as central to all human thinking'. There is no information in the passage that can be used to infer either of the options A, B, C or D. Therefore, option E is the answer.

25. D

'Deterministic lawfulness' here refers to the assertion that it is possible, on the basis of economic principles, to establish a cause effect relationship between events. Option D is in keeping with this and is the correct answer.



Decision Making

26. A

The case does not give any reason to fire or demote Kale. The immediate course of action should be to investigate the accusations against Kale. If the charges are proved correct, Kale should be transferred as he can work effectively when working alone. In order to help Kale better, Marathe can suggest family counselling as a long term solution. Family problems may be affecting Kales work as well but there is nothing in the passage to suggest that. Hence the correct sequence is 2,1,5.

 VIDEO SOLUTION

27. E

The only solution that pertains to family problem (since it deals with family issues) is mentioned in statement 5 of the previous question - "Marathe should suggest Kale to visit a family counsellor." Therefore if Marathe has to agree with Lakhote's opinion of not addressing the personal problems of employees then this solution (statement 5) becomes invalid. Hence the correct answer is option E.

 VIDEO SOLUTION

28. C

Option A is negated because the major concern was Kale's inability to work competently in a team and not his conflicting responsibilities at home and office. Kale was facing personal problems at home and option B states that employees' problems at home affect their performance at work. This strengthens the argument rather than weakening it by citing one more reason to sack Kale. Option D is incorrect because if it holds true then it wipes out the possibility of sending Kale to some other location for work and does leave the management with the

only option of sacking him. Option E is eliminated because it again strengthens the argument as the caselet clearly says that Kale was emotionally unstable. Option C is the correct choice as the report in the journal indicates that most managers find it difficult to work in a group. Even if Kale is sacked and a new manager is hired in his place, it is likely that the new manager may face the same problem (of not being able to work in a group). And hence it will not be a good business decision to sack Kale. Therefore, option C weakens the argument and is the correct answer.

 VIDEO SOLUTION

29. **E**

Appreciation by boss can never be demotivating for an employee. It will rather boost the morale of an employee and motivate him to work to hard. Hence, Vardarajan's declining involvement cannot be attributed to the fact that his bosses appreciated him. Thus, option E is the most unlikely reason for his declining involvement in work. Options A, B, C and D, if true, could be considered as likely reasons for Vardarajan's disinterest in work and hence are incorrect.

30. **A**

The most appropriate combination would be 1, 2, 3. Giving Vardarajan a more challenging assignment would address his lack of interest in work. Transferring him from the Projects department to the Training department could provide him with new opportunities and a change of environment. Asking him to take a vacation for two months would give him a break and a chance to recharge. These steps together would help to motivate Vardarajan and address his lack of interest in work.



31. **B**

Option A is incorrect because promoting Vardarajan will not address the issue at hand. The problem can only be addressed by keeping the performance assessment system transparent and by ensuring impartiality in reviewing performance of each employee. Thus, the correct answer is given by option B. Options C, D and E are negated for the same reason as A i.e. the issue cannot be resolved by addressing Vardarajan's case alone but by fairly judging the performance of all the employees in the organization.

32. **C**

Loss of retail customers does not necessarily mean loss of revenue. It is important to understand that this reduction in retail customers may be by design and an organizational objective. The organization may have been trying to shift its customer base from retail to commercial and their success in doing so can account for the increase in revenue. Hence answer choice C is correct.

33. **A**

The reason why doctors have not embraced the surgery has to be linked to its potential side effects. Answer choice A tells us that the surgery can result in serious complications including death and this is a reason enough for the doctors to not recommend its use. Options B and C tell us the prevalence and effects of obesity which do not explain why doctors are wary of the discussed mode of treatment. Option D talks about how other methods like dieting are not very effective. Option E is irrelevant as it talks about cost and financing, something which the doctor may not be concerned with.

34. **B**

Shambu's initial success was due to the latent anger in the people and Shambu's moral courage to stand up against corruption. Raising income level was not the reason for Shambu's initial success. Also, Shambu became the darling of the media once he was successful. It has not been given that media aided in furthering Shambu's struggle against corruption.

Points 1 and 2 played a pivotal role in Shambu's initial success. Therefore, points 3 and 4 are unlikely to be the reasons for Shambu's initial success and hence, option B is the right answer.

35. **C**

Option B is false as state of Sambhu cannot be decided according to the paragraph.

In the last line of the second paragraph it is mentioned that " He used this opportunity as a platform to spread the message that only citizens with an unblemished character should be allowed to hold a public office." Thus Sambhu was not directly involved in the politics, he just uses the stage to garner support, which might be the reason for decline.

Option D and E are trivial.

Option C is correct



36. E

The minister has stated that the loss is notional and in reality, the projects have benefited everyone. Options C and D talk about actual losses. The minister has not made any statement regarding the loss. Therefore, we can eliminate options C and D. Options A and B talk about notional losses. Again, notional losses should not be a factor deciding whether a project should be implemented or not. We can eliminate options A and B as well. None of the given statements are valid inferences and hence, option E is the right answer.

37. D

From the statements of minister and his answers to the questions raised by the Head of the nation, it is clear that the minister has allocated resources following the rules set by the government and there is no partiality or irregularities have taken place. So, Media has wrongly accused Minister.

Option A is wrong as we can't define a boundary to media.

There is no support to validate claims in option B.

Option C is wrong.

Option D is correct answer.

38. D

The correct answer to this questions should have been 'both corporate managers and the minister'. The head of the nation should not have been accused as he was the one to identify the discrepancy in the allotments. Since the correct answer is not given in the answer choices, XAT picked option D as the correct answer.

39. C

Sales for online = 50% of 60000 = 30000

Remaining sales = 60000-30000 = 30000

The ratio of supermarket sales and book store sales is 1:2

Supermarket sales = $(30000 \times 1)/3 = 10000$

Bookstore sales = $(30000 \times 2)/3 = 20000$

Profit = Sales - Initial costs - Additional costs

Sales and Additional costs can be calculated from the table. But no information about distribution of initial costs among different retails formats. From statement 2 the distribution of initial costs is clear.

Hence both statements will be used to calculate profit for different retail formats.

40. D

Sales from online = 50% of total sales = 50% of 60000 = 30000

Initial costs for online = 50% of total initial costs (Initial cost is allocated in ratio of sales)
= 50% of 39000 = 19500

Additional costs = Red + Yellow + Green + Violet for online)(1)

Total red costs = 5500

Red costs for online = $5500 \times (200/275) = 4000$

Similarly,

Yellow costs for online = $3100 \times (52/100) = 1612$

Green costs for online = $4800 \times (50/80) = 3000$

Violets costs for online = $2400 \times (50/80) = 1500$

Addition costs for online = $4000 + 1612 + 3000 + 1500 = 10112$ From (1)

Profit = Sales - Initial costs - Additional costs = $30000 - 19500 - 10112 = 388$



41. B

Sales from online = 50% of total sales = 50% of 60000 = 30000

The ratio of supermarket sales and book store sales is 1:2

Supermarket sales = $(30000 \times 1)/3 = 10000$

Bookstore sales = $(30000 \times 2)/3 = 20000$

Initial costs for online = 50% of total initial costs

(Initial cost is allocated in ratio of sales)

= 50% of 39000 = 19500

Initial costs for supermarket = $(19500 \times 1)/3 = 6500$
(Initial cost is allocated in ratio of sales)

Initial costs for bookstore = $(19500 \times 2)/3 = 13000$

For online:

Additional costs = Red + Yellow + Green + Violet for online)(1)

Total red costs = 5500

Red costs for online = $5500 \times (200/275) = 4000$

Similarly,

Yellow costs for online = $3100 \times (52/100) = 1612$

Green costs for online = $4800 \times (50/80) = 3000$

Violets costs for online = $2400 \times (50/80) = 1500$

Addition costs for online = $4000 + 1612 + 3000 + 1500 = 10112$ From (1)

Profit = Sales - Initial costs - Additional costs = $30000 - 19500 - 10112 = 388$

For Supermarket:

Additional costs = Red + Yellow + Green + Violet

= $5500 \times (65/275) + 3100 \times (20/100) + 4800 \times (21/80) + 2400 \times (21/80) = 3810$

Profit = $10000 - 6500 - 3810$

= -310

For Bookstore:

Additional costs = Red + Yellow + Green + Violet

= $5500 \times (10/275) + 3100 \times (30/100) + 4800 \times (9/80) + 2400 \times (9/80)$

Profit = $20000 - 13000 - 1940 = 5060$

Supermarket is making least profit for BnC.

42. A

Sales from online = 50% of total sales = 50% of 60000 = 30000

The ratio of supermarket sales and book store sales is 1:2

Supermarket sales = $(30000 \times 1)/3 = 10000$

Bookstore sales = $(30000 \times 2)/3 = 20000$

Initial costs for online = 50% of total initial costs

(Initial cost is allocated in ratio of sales)

= 50% of 39000 = 19500

Initial costs for supermarket = $(19500 \times 1)/3 = 6500$
of sales)

(Initial cost is allocated in ratio

Initial costs for bookstore = $(19500 \times 2)/3 = 13000$

For online:

Additional costs = Red + Yellow + Green + Violet for online)(1)

Total red costs = 5500

Red costs for online = $5500 \times (200/275) = 4000$

Similarly,

Yellow costs for online = $3100 \times (52/100) = 1612$

Green costs for online = $4800 \times (50/80) = 3000$

Violets costs for online = $2400 \times (50/80) = 1500$

Addition costs for online = $4000 + 1612 + 3000 + 1500 = 10112$ From (1)

Profit = Sales - Initial costs - Additional costs = $30000 - 19500 - 10112 = 388$

For Supermarket:

Additional costs = Red + Yellow + Green + Violet

= $5500 \times (65/275) + 3100 \times (20/100) + 4800 \times (21/80) + 2400 \times (21/80) = 3810$

Profit = $10000 - 6500 - 3810$

= -310

For Bookstore:

Additional costs = Red + Yellow + Green + Violet

= $5500 \times (10/275) + 3100 \times (30/100) + 4800 \times (9/80) + 2400 \times (9/80)$

Profit = $20000 - 13000 - 1940 = 5060$

Bookstore gives highest profit for BnC.

43. C

As the probability of getting incorrect answers is quite high in comparison to getting right answers for the students of the class in the test, in case students sitting next to each other go for cheating, then probability of them getting same incorrect answers for multiple questions is quite high. Therefore, by tracking pattern of incorrect answers, it is possible to detect whether cheating took place or not.

44. E

Probability of the first student of the group of three getting 4 incorrect answer of 1st question = $\frac{4}{5}$

Probability of the second student of the group getting the same incorrect answer of 1st question as that of the first student = $\frac{4}{5} \times \frac{1}{5}$

Probability of the third student of the group getting the same incorrect answer of 1 st question as that of the first student = $\frac{4}{5} \times \frac{1}{5}$

Therefore, the probability of all the three getting same incorrect answer of 1st question = $\frac{4}{5} \times \frac{4}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{1}{5} = \frac{4^3}{5^5}$

Hence, the probability = $\left(\frac{4^3}{5^5}\right)^4$

45. **E**

Option A: we get that $A > B > C > D$ and $C > 2$

Option B: we get A is the largest, $C > 2$, $A > 4$ and every section has a different number of students.

Option C: we get that $A > B > C > D$, $C > 2$ and $A * B * C * D = 10n$

Option D: we get $A > B > C > D$, $C > 2$ and $A * B * C * D = 24n$

Option E: we get A is the largest, D is the smallest, $C > 2$, $A * B * C * D = 24n$, and each section has a different number of students.

Statement 9 tells us that each section contributes a different number of students and thus this statement is a must in figuring out each section's strength.

Thus, options A, C and D are eliminated. From option B we cannot figure out any value since we don't have any relation between their values. Hence, option B is eliminated.

Option E tells us about the largest and smallest section, also provides a relation between all their values which can be used to find out possible solutions and also tells us that each section contributes a different number of students which can be used to eliminate the undesirable results.

Thus, option E is the answer.



XAT Syllabus PDF



46. **A**

We have been given that A, B, C, D and E are not holding the 1, 2, 3, 4 and 5 ranks, respectively.

Also that two of the positions in the arrangement DAECB are correct and the other two follow their immediate predecessors.

The only arrangement satisfying both these conditions is EDACB.

Thus, option A is the correct answer.

47. **C**

The caselet presents a complicated problem making it difficult to find out the guilty. The best way to proceed in cases like these is to launch an investigation into the matter so as to find out that who is at fault. Thus, the most appropriate course of action to be followed is presented by option C - Vijya should constitute a fact finding committee to investigate the matter and then decide about the next step to be taken. Options A and B are inappropriate because these steps could be taken only after finding out the person at fault. Option D is eliminated as involving the corporate recruiters in the matter will only aggravate the issue and bring a bad name to the college. An effort should be made to try and solve the problem internally rather than publicizing it.

48. **D**

Option D is the correct answer. It is advisable to start from the scratch and shortlist candidates from the fresh set if bio-data. Since the fresh set contains the verified grades of the students so there are no chances of any discrepancy this time. Option A is incorrect as it is an extreme step and should not be taken when there is an option of screening the candidates again. Option B can be negated because the HR managers's contact may not have the correct information. Furthermore it does not address the problem of recruitments from that college. It is unethical for an HR manager to recruit candidates even after knowing that that the information presented in their bio-data's is not authentic. This will show his disloyalty towards the organization. Option E is incorrect as it reduces the pool of students to be recruited from.

49. **B**

On being asked about information regarding pay packages of graduating students, the senior does not reveal much information and rather cautions you not to believe in any sort of rumors. He also said that for any further information the college authorities should be contacted directly. Thus option B presents the most suitable course of action for seeking further information about placements. Option A is incorrect because the senior has already warned you that believing in rumors will do you no good. The information that you get by networking with students may not be authentic. Option C can be eliminated because contacting the HR managers is beyond the purview of discussion held between the potential entrant and the senior. Option D is incorrect because the senior has advised to contact the student body and not any individual representing the body. Approaching the President directly may not prove to be of any help. Option E is negated for the same reason as option A.

50. D

Option A is inappropriate as it will be unprofessional on Vijya's part to suppress this information by not paying heed to the phone call. Options B and C are negated because it is unethical to shrug off the responsibility by absenting oneself from work or by putting it on someone else. Option E is eliminated as it is not feasible to constitute a committee for every issue. Furthermore, this will only prolong things and will delay the matter. Option D is the correct answer. Since the matter cannot go unreported so it is advisable to consult the Director (Vijya's senior) and seek his opinion on the issue.



Quantitative Ability and DI

51. D

Mean of the six numbers = 15

So, the sum of the numbers = $15 \times 6 = 90$

As the median is 18, the mean of middle two numbers must be 18 and thus, their sum must be 36.

Also, the mode is a number less than 18. So, the mode must be appearing as the first and the second number of the six given integers, when arranged in ascending order. To maximize the largest integers, the mode must be equal to 1.

Therefore, out of the six integers, two are 1 and 1.

For the middle two numbers whose sum is 36, we cannot have 18 and 18 because then we will have two modes which is inappropriate as per the question.

So, the middle numbers must be 17 and 19.

The fifth integer can be 20.

Maximum possible value of the largest integer = $90 - (1 + 1 + 17 + 19 + 20) = 32$

Hence, option D is the correct answer.

 VIDEO SOLUTION

52. E

We know that Ramesh bought 6 fruits in total.

If the number of apples is 1, then the number of oranges required to get an equal amount of juice will be 0.

Therefore, we can eliminate this case.

If the number of apples is 2, then the number of oranges required to get an equal amount of juice will be 1. We know that Ramesh had 1 more orange than needed. The total number of fruits in this case is $2 + 1 + 1 = 4$.

Therefore, we can eliminate this case too.

If the number of apples is 3, then the number of oranges required to get an equal amount of juice will be 2. We know that Ramesh had 1 more orange than needed. The total number of fruits in this case is $3 + 2 + 1 = 6$. This satisfies the condition.

The quantity of juice from 3 apples is equal to the quantity of juice from 2 oranges.

Therefore, the proportion of apple juice in the initial mixture is $2/(2+3) = 2/5 = 40\%$. (2+3 is used since we are finding the quantity of juice. 2 denotes the quantity of juice obtained from 3 apples)

Therefore, option E is the right answer.

53. **B**

Sarah has 5 friends and each of her 5 friends have 25 friends.

The 5 friends of Sarah have Sarah as one of their 25 friends. Therefore, apart from Sarah the five friends will have 24 friends each.

If no 2 friends know each other, then Sarah will have to invite $5 \times 24 + 5 = 120 + 5 = 125$ persons.

Sarah knows that at least 2 of her friends are connected to each other.

Let the 5 friends be A, B, C, D, and E. Let us consider that A and B are friends.

A will be counted twice (first as one among the 5 persons and second as one of the friends of B).

B will also be counted twice (first as one among the 5 persons and second as one of the friends of A).

Subtracting these 2 friends, we can infer that the maximum number of persons that Sarah could have invited to the party is 123.

The minimum number of friends that Sarah could have invited is obtained when all the friends are connected to each other and their friends are also the same. In this case, the five friends will have 20 common friends, will be friends with each other (4) and Sarah (1).

The minimum number of persons Sarah could have invited to the party = $20 + 5 = 25$.

Option B is more appropriate than option C.

Therefore, option B is the right answer.

54. **D**

As it took Prof. Mandal 30 minutes for round trip to the market by auto, we can infer that the auto takes 15 minutes for one way trip.

When he walks one way and return by auto, it took him 90 minutes.

We know the auto would have taken 15 minutes while returning.

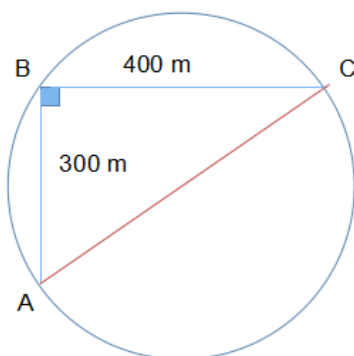
So, the professor takes $(90 - 15) = 75$ minutes one way while walking to the market.

So, for a round trip by walking, he will take 2×75 minutes = 150 minutes.

Hence, option D is the correct.

 VIDEO SOLUTION

55. **A**



The fish travels North for 300 m and on hitting the edge, travels East for 400 m. The directions North and East are perpendicular to each other. On connecting the initial and final positions of the fish in the tank, we get a right-angled triangle.

The line AC subtends an angle of 90 degree on the circumference. Therefore, AC must be the diameter of the pond.

Applying Pythagoras theorem, we get,

$$AB^2 + BC^2 = AC^2$$

$$AC^2 = 300^2 + 400^2$$

$$AC = 500 \text{ m.}$$

=> Radius of the pond = 250 m.

$$\text{Area of the pond} = \pi * r^2$$

$$\Rightarrow \text{Area} = 62500\pi m^2.$$

Therefore, option A is the right answer.



56. E

Rs. 15 lakhs is to be divided equally.

In the case of the younger son,

Principal = Rs. 750000, time = 9 years and Interest = Rs. 1350000

$$\text{Rate of interest} = \frac{1350000 * 100}{750000 * 9} = 20\%$$

In the case of the elder son,

Principal = Rs. 750000, time = 6 years and Interest = Rs. 1350000

$$\text{Rate of interest} = \frac{1350000 * 100}{750000 * 6} = 30\%$$

Hence, option E is the correct answer.

VIDEO SOLUTION

57. A

Let the total time required be 12T (LCM of 2, 3 and 4 for the ease of calculation)

No. of pages read by Albela in 12T minutes at 2 minute per page = 6T

No. of pages read by Bob in 12T minutes at 3 minute per page = 4T

No. of pages read by Chulbul in 12T minutes at 4 minute per page = 3T

Total no. of pages read by all three of them = (6T + 4T + 3T) = 13T

According to the question, total pages = 78

So, 13T = 78

Or, T = 6

Therefore, no. of pages read by Bob = 4T = 24

Hence, option A is the correct answer.

VIDEO SOLUTION

58. C

For the first 2 km, fare = Rs. 20

For the next 8 km, fare = Rs. 5 * 8 = Rs. 40

For the next 4 km, fare = Rs. 8 * 4 = Rs. 32

Total fare up to 14 km = Rs. 20 + Rs. 40 + Rs. 32 = Rs. 92

Total fare paid by Sardar Khan = Rs. 102

Money spent on bullock cart = Rs. 102 - Rs. 92 = Rs. 10

Charge for bullock cart = Rs. 2/km

Distance travelled by bullock cart = 5 km

Total distance travelled by Sardar Khan = (14 + 5) km = 19 km

Hence, option C is the correct answer.

59. E

The given expression can be written as $\frac{a^2+a+1}{a} * \frac{b^2+b+1}{b} * \frac{c^2+c+1}{c} * \frac{d^2+d+1}{d} * \frac{e^2+e+1}{e}$.

$$\frac{a^2+a+1}{a} = a + \frac{1}{a} + 1$$

We know that for positive values, $AM \geq GM$.

$$\frac{a+\frac{1}{a}}{2} \geq \sqrt{a * \frac{1}{a}}$$

$$a + \frac{1}{a} \geq 2$$

The least value that $a + \frac{1}{a}$ can take is 2.

Therefore, the least value that the term $a + \frac{1}{a} + 1$ can take is 3.

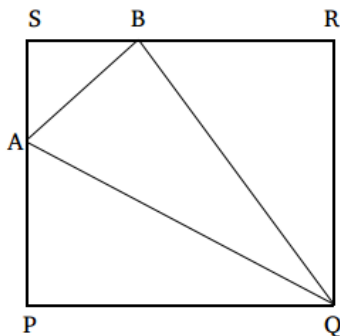
Similarly, the least value that the other terms can take is also 3.

=> The least value of the given expression = $3 * 3 * 3 * 3 * 3 = 243$.

Therefore, option E is the right answer.

▶ VIDEO SOLUTION

60. C



$$PQ = QR = RS = SP = 6 \text{ cm}$$

$$PA = 2AS \text{ and } PA + AS = 6 \text{ cm}$$

$$\Rightarrow PA = 4 \text{ cm and } AS = 2 \text{ cm}$$

$$\text{Similarly, } RB = 2BS \text{ and } RB + BS = 6 \text{ cm}$$

$$\Rightarrow RB = 4 \text{ cm and } BS = 2 \text{ cm}$$

$$\text{Area of } \triangle ABQ = \text{area of PQRS} - \text{area of } \triangle APQ - \text{area of } \triangle RBQ - \text{area of } \triangle ASB$$

$$= (36 - 12 - 12 - 2) \text{ sq. cm}$$

$$= 10 \text{ sq. cm}$$

Hence, option C is the correct answer.

▶ VIDEO SOLUTION



61. B

Let us analyse all numbers from 100 to 200 (both excluded).

$$\text{Total numbers when unit digit is 2} = 10 \{102, 112, 122, \dots, 192\}$$

$$\text{Total numbers when tens digit is 2} = 10 \{120, 121, 122, \dots, 129\}$$

We can see that there is exactly one number {122} which has both tens and unit digit as 2. So total unique numbers from 100 to 200 (both excluded) which contain at least one 2 digit = $10 + 10 - 1 = 19$

From 200 to 299 (both included), all numbers have at least one digit as 2. Total such number = 100.

From 300 to 400, 400 to 500, 500 to 600, 600 to 700 and 700 to 800 we will have 19 numbers for each pair with at least one 2 digit.

Therefore, total whole numbers between 100 and 800 contain the digit 2 = $100 + 19 * 6 = 214$.

 VIDEO SOLUTION

62. C

The product of 2 numbers A and B is maximum when $A = B$.

If we cannot equate the numbers, then we have to try to minimize the difference between the numbers as much as possible.

pq will be maximum when $p=q$.

qr will be maximum when $q=r$.

pr will be maximum when $r=p$.

Therefore, p, q, and r should be as close to each other as possible.

We know that p, q, and r are integers and $p+q+r=10$.

$\Rightarrow$ p, q, and r should be 3, 3, and 4 in any order.

Substituting the values in the expression, we get,

$$pq+qr+pr+pqr = 3*3 + 3*4 + 3*4 + 3*3*4$$

$$= 9 + 12 + 12 + 36$$

$$= 69$$

Therefore, option C is the right answer.

 VIDEO SOLUTION

63. E

As the number is between 10 and 100 and 100 cannot be the number we are looking for, we can assume the number to be of two-digits.

Let the number be xy.

According to the question, for the number to be interesting

$$x + y + xy = 10x + y$$

On solving, we get

$$xy = 9x$$

$$\text{or, } x(9 - y) = 0$$

x cannot be 0, because we need a number greater than or equal to 10.

$$\text{So, } 9 - y = 0$$

$$\Rightarrow y = 9$$

For all the numbers whose unit digit is 9 will be an interesting number.

So, the numbers are 19, 29, 39, 49,99

There are 9 such numbers out of 91 total numbers between 10 and 100 including both.

$$\text{Required fraction} = \frac{9}{91} = 0.0989$$

As this is not given in any of the options, the answer will be "none of the above".

Hence, option E is the correct answer.

 VIDEO SOLUTION

64. B

Let '100x' be the number of students who joined XCRI last year.

Let 'a', 'b', 'c' and d be the number of students who play 1 game, 2 games, 3 games and 4 games respectively.

Therefore,

$$a+b+c+d = 100x \dots (1)$$

$$a+2b+3c+4d = 70x+75x+80x+85x$$

$$a+2b+3c+4d = 310x \dots (2)$$

By equation (2) - (1)

$$b+2c+3d = 210x$$

We have to minimize 'd' for that we have to maximize c. But $c \leq 100x$

At $c_{max} = 90x$, $d_{min} = 10x$

Therefore, we can say that the minimum percentage of students who play all four games = 10%.

65. D

$$p^q = q^p.$$

It has been given that $q = 9p$.

Substituting, we get,

$$p^{9p} = (9p)^p$$

$$(p^p)^9 = 9^p * p^p$$

$$\Rightarrow (p^p)^8 = 9^p$$

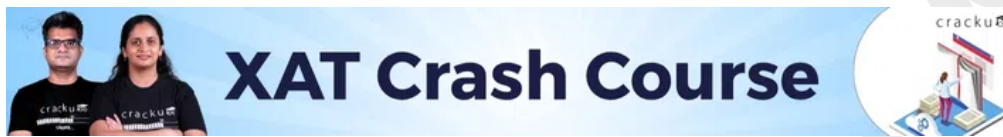
$$p^{8p} = 9^p$$

Raising the power to $\frac{1}{p}$ on both sides, we get,

$$p^8 = 9$$

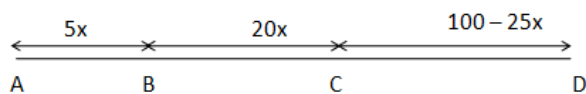
$$p = \sqrt[8]{9}.$$

Therefore, option D is the right answer.



66. A

Let 'A' be the point from where all started and D is the destination. At 'x' hours both Ram and Hari reached point C, Hari got off and Ram turned back to pick up Shyam. At the same instant, Shyam was at point B.



Let 'T' is the total amount of time taken by all three to reach point D.

For Hari,

$$T = x + \frac{100 - 25x}{5} \dots (1)$$

For Ram,

$$T = x + \frac{20x * 5/6}{25} + \frac{20x * 5/6 + 100 - 25x}{25} \dots (2)$$

By equating (1) and (2),

$$x + \frac{100 - 25x}{5} = x + \frac{20x * 5/6}{25} + \frac{20x * 5/6 + 100 - 25x}{25}$$

$$\Rightarrow x = 3 \text{ hours.}$$

$$\text{Therefore, time taken by all to reach destination} = x + \frac{100 - 25x}{5}$$

$$\Rightarrow 3 + \frac{100 - 25 * 3}{5}$$

$$\Rightarrow 3 + 5$$

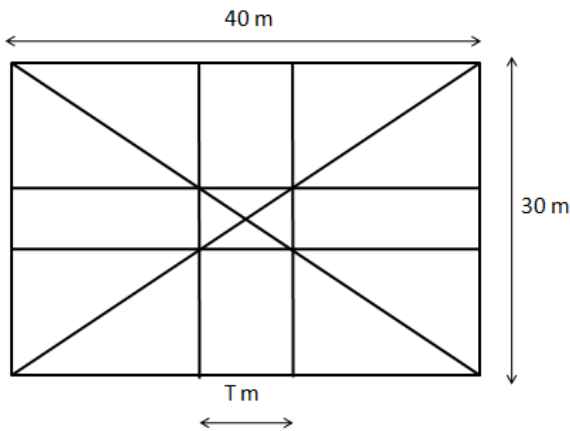
$$\Rightarrow 8 \text{ hours}$$

Hence, we can say that option A is the correct answer.

VIDEO SOLUTION

67. A

Let us draw the diagram according to the information given in the questions.



Let us assume that 'T' is the width of the road as shown in the diagram.

$$\text{Total area covered by road} = 30 * T + 40 * T - T^2$$

Also it is given that the mayor wants that the area of the two roads to be equal to the remaining area of the park.

$$\Rightarrow 30 * T + 40 * T - T^2 = \frac{30 * 40}{2}$$

$$\Rightarrow T^2 - 70T + 600 = 0$$

$$\Rightarrow (T - 10)(T - 60) = 0$$

$$\Rightarrow T = 10 \text{ or } 60 \text{ m. } T \neq 60 \text{ m as the width of park is 30 m only.}$$

Therefore, we can say that $T = 10$ meters. Hence, option A is the correct answer.

68. B

It has been given that Arun's task is to complete as many tasks as possible and then to focus on completing tasks based on their priority.

Arun has 10 days with him. Except 1 task, all other tasks require at least 3 days to be completed. Therefore, Arun can complete a maximum of 3 tasks.

Arun can complete T1, T2, and T5 or T1, T3, and T4 or T1, T3, and T5 or T2, T3, and T5 or T1, T4, and T5.

In all these combinations, Arun would have completed 2 higher priority tasks in the combination T1, T2, and T5.

Therefore, Arun should complete T1, T2, and T5 and hence, option B is the right answer.

69. E

According to the data given in previous question, Arun have 10 days to do a background check, as he have to enter within 10 days of receiving permit.

The number of tasks that can be completed in this case is also 3.

Arun can complete T1, T2, and T5 or T1, T3, and T4 or T1, T3, and T5 or T2, T3, and T5 or T1, T4, and T5.

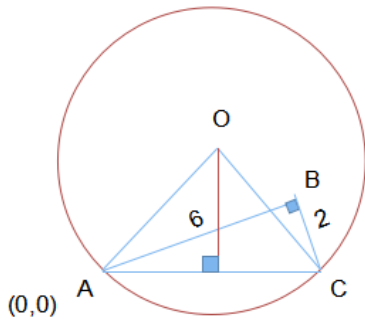
In this case, the time required to complete T1, T2, and T5 exceeds 10 days. The same is the case with T1, T2, and T3.

Among the given options, both options D and E are possible cases.

T5 is a higher priority task than T3. Therefore, Arun should try to complete T1, T4, and T5.

Therefore, option E is the right answer.

70. A



We know that ABC is a right angled triangle.

$$\Rightarrow AC = \sqrt{6^2 + 2^2}$$

$$AC = \sqrt{40}$$

$$AC = 2 * \sqrt{10}$$

Let the coordinates of A be (0,0)

We know that the radius of the circle, $OA = \sqrt{50}$ cm

Let OD be the height of the triangle AOC.

$$AC/2 = * \sqrt{10}$$

By applying Pythagoras theorem, we get,

$$\Rightarrow \text{The height(y) of the point O} = \sqrt{50 - 10}$$

$$\text{The height(y) of the point O} = \sqrt{40}$$

$$\Rightarrow \text{Coordinates of point O} = (\sqrt{10}, \sqrt{40})$$

Area of triangle ABC = $0.5 * 6 * 2 = 6$ square units.

Let the height of triangle ABC be h.

$$0.5 * h * AC = 6$$

$$h * AC = 12$$

$$h * 2 * \sqrt{10} = 12$$

$$h = \frac{6}{\sqrt{10}}$$

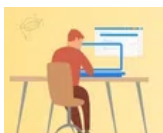
$$\begin{aligned} \text{X-coordinate of point B} &= \sqrt{6^2 - \frac{36}{10}} \\ &= \sqrt{32.4} \text{ cm} \end{aligned}$$

$$\text{Distance between points O and B} = \sqrt{(\sqrt{10} - \sqrt{32.4})^2 + (\sqrt{40} - \frac{6}{\sqrt{10}})^2}$$

Expanding, we get,

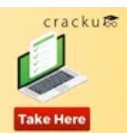
Square of the distance between points O and B = 26 cm.

Therefore, option A is the right answer.





XAT Sectional Tests



71. E

There are 6 cards and 2 out of the 6 cards are kings.

Number of ways of selecting 2 cards = $6C2 = 15$ ways.

Number of ways in which 2 cards can be selected such that both of them are King = $2C2 = 1$

Number of ways in which 2 cards can be selected such that exactly one of them is a King = $2C1 * 4C1 = 8$

$$\Rightarrow a = (1+8)/15 = 9/15$$

$$b = 1 - (9/15) = 6/15$$

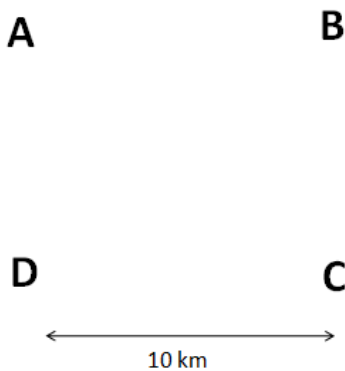
$$a/b = 9/6 = 1.5$$

$$1.5 > 1.25$$

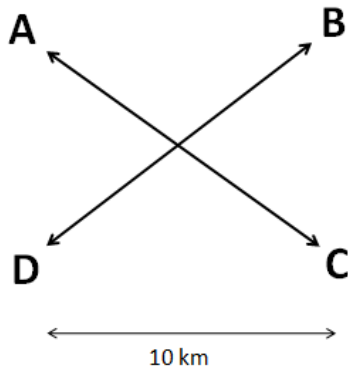
Therefore, option E is the right answer.

72. C

We can draw the towns as shown in the diagram below.



For the minimum length the Government should diagonal roads. The length of road will be same as the length of a diagonal of a square whose side length is 10 km.



Length of one diagonal = $\sqrt{2} * 10 = 1.414 * 10 = 14.14$ km

Therefore, total length of both the roads = $2 * 14.14 \approx 28.30$ km.

73. E

$$(xxx)_b = x^3$$

$$\Rightarrow xb^2 + xb + x = x^3$$

$$\Rightarrow b^2 + b + 1 = x^2$$

On substituting $b=1$, and $b=2$, we get x^2 as 3, and 7. Since 3 and 7 are not perfect squares, we can infer that no number satisfies the given condition. Therefore, option E is the right answer.

74. A

$$x^5 - 1 = (x - 1)(x^4 + x^3 + x^2 + x + 1)$$

$$x^5 - 1 = (x - 1)f(x)$$

$$x^5 = 1 + (x - 1)f(x)$$

$$f(x^5) = (1 + (x - 1)f(x))^4 + (1 + (x - 1)f(x))^3 + (1 + (x - 1)f(x))^2 + (1 + (x - 1)f(x)) + 1$$

On dividing by $f(x)$, every term will leave 1 as the remainder. Therefore, the remainder when $f(x^5)$ is divided by $f(x)$ is $1 + 1 + 1 + 1 + 1 = 5$.

Therefore, option A is the right answer.

75. C

Let us evaluate the statements one by one:

I: 103 and 7 are the only prime factors of 1000027

On successively dividing 1000027 by 103 and 7, we get 1387 as the answer.

1387 is divisible by 19.

Therefore, statement I is false.

II: $\sqrt[6]{6!} > \sqrt[7]{7!}$

Raising the power to 42 on both sides, we get,

$$[6!]^7 > [7!]^6$$

$$6! * [6!]^6 > 7^6 * [6!]^6$$

7^6 is greater than $6!$.

Therefore, statement II is false.

III: It has been given that the person travels one-half of the journey at x kmph.

Let us assume the distance to be '2d'.

Let us assume the average speed to be 2d and check for the feasibility.

Let the speed at which the person travels the other half of the journey be y.

$$d/x + d/y = 2d/2x$$

$$d/x + d/y = d/x$$

$$\Rightarrow d/y = 0 \text{ or } y \text{ tends to infinity.}$$

Therefore, such a scenario is not possible and hence, statement III is true.

Only statement III is true. Therefore, option C is the right answer.

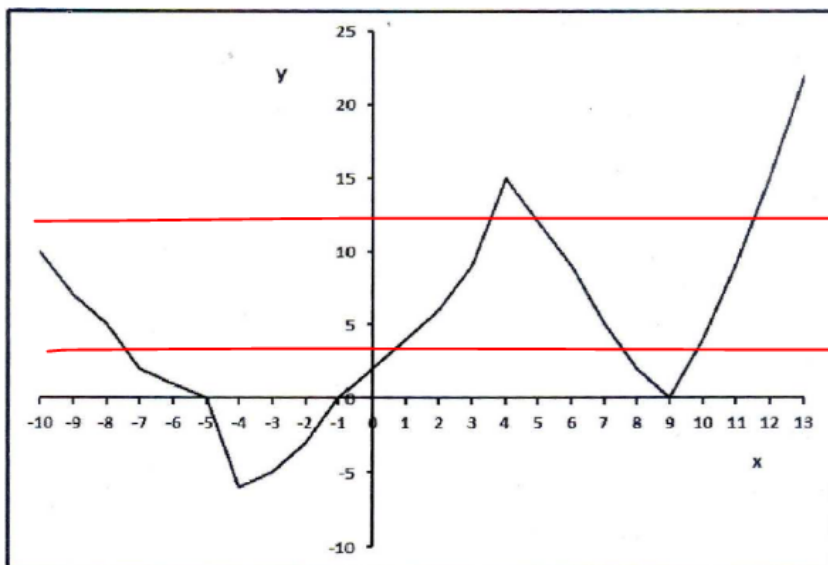


76. C

It has been given that $f(f(x)) = 15$.

From the graph, we can see that the value of $f(4) = 15$ and $f(12) = 15$

Therefore, $f(x)$ can be 4 or 12.



When $f(x) = 4$, x can take 4 values

When $f(x) = 12$, x can take 3 values.

Therefore, there are $4+3 = 7$ solutions in total.

Therefore, option C is the right answer.

77. E

Current account balance = Current account balance percentage of GDP * GDP

Current account balance in 2010 = 3.268% of 73555.34 = 2403.7885

Current account balance in 2005 = 1.272% of 35662.2 = 453.623

Ratio = $2403.7885 / 453.623 = 5.3$ (approximately).

Therefore, option E is the right answer.

78. C

The table provides details only regarding the % change in the imports and exports. We do not have any detail regarding the base value. Therefore, we cannot find out the absolute values and hence, statements 1 and 2 cannot be verified.

The percentage change of exports and imports over the given period can be found out. Among the given options, option C is the only viable option now. Since none of the above is not present among the options, option C must be the right answer.

To actually compare the growth rate of export and import. we can multiply the rates to find the net change and then compare.

79. A

Population = GDP / GDP per capita

In 2006, GDP = Rs. 41159.73 billion

GDP per capita = Rs. 36,553.93

=> Population = $41159.73 \times 1000 / 36553.93$
= 1126 million.

From the table, we can see that 8.9% of the total population was unemployed in 2006.

=> Number of unemployed people = $0.089 \times 1126 = 100$ million (approx)

Therefore, option A is the right answer.

80. D

Except 7777, all other trains take at least 2 hours to cover the distance between the stations HHH and NNN.

Therefore, option D is the right answer.



XAT Mock Test

Mocks analyzed by Sayali Ma'am (All-India XAT topper)



81. A

In the graph, the line with the steepest slope should be preferred to travel in the shortest possible time.

The line representing 8800 completes within 2 hour slot and has the steepest slope among the given options.

Therefore, option A is the right answer.

82. D

Only 3 trains - 4444, 5555, and 8888 reach HHH before 9 AM. Among these 3 trains, 8888 reaches just before 9 AM and hence, it is the best train to reach HHH from AAA. 8800 reaches HHH at 6 PM exactly. Therefore, it is the best train to travel from HHH to NNN. Therefore, option D is the right answer.

83. D

Using the data from both the tables available, we can create the table for away matches as given below:

X	M	W	D	L	GF	GA	GD	Points	Rank
AS	2	1	1	0	2	0	2	4	1
SW	1	1	0	0	5	0	5	3	2
CH	1	1	0	0	2	0	2	3	3
EV	2	1	0	1	3	3	0	3	4
MU	2	1	0	1	3	3	0	3	5
ST	2	0	2	0	3	3	0	2	6
MC	1	0	1	0	2	2	0	1	7
WB	1	0	1	0	1	1	0	1	8
NC	1	0	0	1	0	2	-2	0	9
WH	1	0	0	1	0	3	-3	0	10
FU	2	0	0	2	2	6	-4	0	11

Thus, the team with rank 2 is SW.

Hence, option D is the answer.

84. **D**

Using the data from both the tables available, we can create the table for away matches as given below:

X	M	W	D	L	GF	GA	GD	Points	Rank
AS	2	1	1	0	2	0	2	4	1
SW	1	1	0	0	5	0	5	3	2
CH	1	1	0	0	2	0	2	3	3
EV	2	1	0	1	3	3	0	3	4
MU	2	1	0	1	3	3	0	3	5
ST	2	0	2	0	3	3	0	2	6
MC	1	0	1	0	2	2	0	1	7
WB	1	0	1	0	1	1	0	1	8
NC	1	0	0	1	0	2	-2	0	9
WH	1	0	0	1	0	3	-3	0	10
FU	2	0	0	2	2	6	-4	0	11

We have to find the number of teams with 0 or 1 point. From the table, we can see that there are 5 such teams.

Thus, option D is the answer.

85. **A**

Using the data from both the tables available, we can create the table for away matches as given below:

X	M	W	D	L	GF	GA	GD	Points	Rank
AS	2	1	1	0	2	0	2	4	1
SW	1	1	0	0	5	0	5	3	2
CH	1	1	0	0	2	0	2	3	3
EV	2	1	0	1	3	3	0	3	4
MU	2	1	0	1	3	3	0	3	5
ST	2	0	2	0	3	3	0	2	6
MC	1	0	1	0	2	2	0	1	7
WB	1	0	1	0	1	1	0	1	8
NC	1	0	0	1	0	2	-2	0	9
WH	1	0	0	1	0	3	-3	0	10
FU	2	0	0	2	2	6	-4	0	11

We have to find the team with the highest difference between home and away rank.

AS: pos = $|12 - 1| = 11$

WB: pos = $|1 - 7| = 6$

WH: pos = $|2 - 10| = 8$

MC: pos = $|4 - 7| = 3$

SW: pos = $|5 - 2| = 3$

Thus, the team with the highest value of pos is AS

Thus, option A is the correct answer.



86. **B**

Using the data from both the tables available, we can create the table for away matches as given below:

X	M	W	D	L	GF	GA	GD	Points	Rank
AS	2	1	1	0	2	0	2	4	1
SW	1	1	0	0	5	0	5	3	2
CH	1	1	0	0	2	0	2	3	3
EV	2	1	0	1	3	3	0	3	4
MU	2	1	0	1	3	3	0	3	5
ST	2	0	2	0	3	3	0	2	6
MC	1	0	1	0	2	2	0	1	7
WB	1	0	1	0	1	1	0	1	8
NC	1	0	0	1	0	2	-2	0	9
WH	1	0	0	1	0	3	-3	0	10
FU	2	0	0	2	2	6	-4	0	11

From the above table, we can see 6 such values: 0, 2, 5, -2, -3 and -4.

Thus, option B is the answer.

General Awareness

87. **A**

Reliance Industries, IOC, TATA steel, and TATA motors featured in the fortune 500 list in 2012. Therefore, option A is the right answer.

88. **A**

The monetary policy is framed by the Reserve Bank of India. The fiscal policy is framed by the ministry of Finance. Therefore, option A is the right answer.

89. **C**

CPI and WPI are price indices used to keep track of inflation.

CPI is the abbreviation of Consumer Price index. It calculates the average price paid by the consumer to the shopkeepers.

WPI is the abbreviation of Wholesale price index. It calculates the price paid by the manufactures and wholesalers in the market.

Therefore, option C is the right answer.

90. **D**

Davos is a town in Switzerland where the annual congregation of the world economic forum takes place. Therefore, option D is the right answer.



XAT Syllabus PDF



91. **E**

The UN set 8 development goals in 2000 to be accomplished by 2015. These goals are known as the millennium goals.

The goals are as follows:

1. Eradicate extreme poverty and hunger.
2. Achieve universal primary education.
3. Promote gender equality
4. Reduce child mortality
5. Improve maternal health
6. Combat HIV and Malaria
7. Ensure environmental sustainability
8. Develop global partnerships for development

Only option E captures these points. Therefore, option E is the right answer.
Millennium development goals were replaced by sustainable development goals in 2015.

92. **D**

When a country is unable to pay its debts, it is said to have been hit by the sovereign debt crisis. In Europe, Greece filed for bankruptcy. Spain, Iceland, and Portugal are also affected by the sovereign debt crisis. Together, these countries are known by the acronym PIGS. Therefore, option D is the right answer.

93. **B**

The top oil producing countries in the world are the United States, Saudi Arabia, Russia, China and Canada. Option B contains 4 of the 5 top oil producing countries in the world. Therefore, option B is the right answer.

94. **E**

The 2012 slide in Indian rupee was caused due to a huge trade deficit, low growth and high inflation, low capital inflows and increasing current account deficit. High employment rate was not one of the reasons for the slide of rupee in 2012. Therefore, option E is the right answer.

95. **A**

IMF and World bank are known as the Bretton woods twins. The decision to form both the institutes arose out of the conference held at Bretton woods on July 1944. Therefore, option A is the right answer.



96. **D**

The Earth Summit, also known as the United Nations conference on environment and development, is held every year. The Earth Summit in 2012 was held in Rio De Janeiro. All the given statements are true. Therefore, option D is the right answer.

97. **B**

France (74.8%), Sweden (38.1%), Korea (30.4%), and Ukraine (46.2%) meet more than 25% of their energy demands using nuclear power.

The US meets only 19% of its energy needs using nuclear power. India meets less than 3% of its energy needs using nuclear power.

England meets 18.1% of its energy needs using nuclear power. Therefore, option B is the right answer.

98. **A**

The options given give the countries and their respective stock exchanges or indices.

Shenzhen stock exchange is located in China.

Both BSE and NSE are located in India.

SGX stands for Singapore Exchange Limited.

DAX is the index of the top 30 German companies.

All the exchanges are correctly matched in option A.

NASDAQ is a stock exchange in the US. FTSE is the index of the 100 countries listed in the London Stock Exchange. Therefore, we can eliminate options B, C, D, and E. Therefore, option A is the right answer.

99. **E**

Ramon Magsaysay award is awarded by the Philippine Government to acknowledge outstanding persons across various fields. Harish Hande was awarded Magsaysay award in 2011. He founded SELCO, a solar energy company, in 1995. The company has made solar lights and low smoke cookers accessible to the poor. Therefore, option E is the right answer.

100. **E**

GAAR stands for General anti-avoidance rule. The rule was established under chapter X-A of Income tax act, 1961. The rule was enacted to curtail tax avoidance. Therefore, option E is the right answer.



101. **C**

Dr. Manmohan Singh was the chairman of the planning commission in 2013. Usually, the prime minister served as the chairman of the planning commission. The planning commission was replaced with NITI Aayog in 2015. Therefore, option C is the right answer.

102. **E**

Palme d'Or is awarded in the Cannes film festival for films. It is considered the highest prize awarded in the festival.

Grammy award is awarded to acknowledge notable contributions in music.

Cannes lions award is awarded for advertisements.

Emmy awards is awarded for television works.

Booker prize is awarded for literature.

Only option E correctly matches the awards with the fields. Therefore, option E is the right answer.

103. **B**

Kindle is the e-book reader released by Amazon.

Angry birds is a product of Rovio.

Wii is the video game console released by Nintendo.

Nook is the e-book reader released by Barnes and Noble.

Castle Ville is an online game released by Zynga.

Adsense is a product of Google.

All the products and the companies are correctly matched only in option B and hence, option B is the right answer.

104. **D**

Excise duty is an indirect form of taxation levied on goods and services by the central government. Excise duty on certain items such as alcohol is collected by the state government. All the given statements are true regarding Excise duty. Therefore, option D is the right answer.

GST has subsumed excise duty in most of the places where excise duty was levied.

105. **D**

Among the given items, India exports gems and jewellery the most. Gems and Jewellery contribute the highest to India's exports. Therefore, option D is the right answer.



106. **C**

Hydrochloric acid, secreted by the lining of stomach, helps in killing the microbes by providing an acidic medium. Therefore, option C is the right answer.

107. **A**

Sushil Kumar is from Haryana.

Mary Kom is from Assam.

Saina Nehwal is from Andhra Pradesh.

Vijay Kumar is from Mhow, Madhya Pradesh.

Yogeshwar Dutt is from Haryana.

Only option A correctly matches the players to the place they belong. Therefore, option A is the right answer.

108. **A**

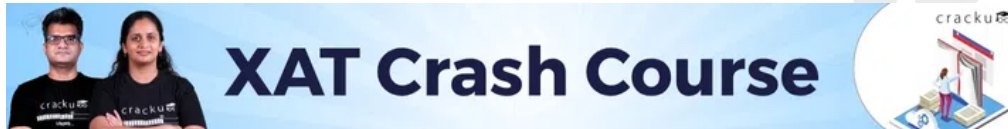
Tunisia, Egypt, Libya, Algeria, Morocco, and Sudan underwent political turmoil between 2011 and 2013. Therefore, option A is the right answer.

109. **E**

Mr. Rajat Gupta, the former chief of McKinsey, is one of the founders of ISB. Therefore, option E is the right answer.

110. **E**

China was the largest producer of cars followed by the US, Japan, Germany, and Korea in that order. Therefore, option E is the right answer.



111. **D**

According to the Economic Survey of India 2011-12, the share of manufacturing sector in the country's GDP lied between 14-16%. Therefore, option D is the right answer.

112. **A**

Amartya Sen was awarded Nobel prize for welfare economics.

Joseph Stiglitz was awarded Nobel prize in economics in 2001 for the analyses of markets with asymmetric information.

Oliver Williamson was awarded the Nobel prize in 2009.

Therefore, option A is the right answer.

113. **C**

In July 2012, violence broke out in the Manesar plant of Maruti Suzuki. The incident left a senior HR official dead. Therefore, option C is the right answer.

114. **C**

One unit of carbon permit is used to refer to one metric tonne of carbon dioxide emission. Therefore, option C is the right answer.